Prog. Aerospace Sci. Vol. 27, pp. 145–200, 1990
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0376–0421/90 $0.00 + 0.50
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SELF-EXCITED OSCILLATIONS IN INTERNAL TRANSONIC FLOWS

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(*Received 26 March 1990*)

Abstract—Compressible internal flows in a velocity regime close to Mach-number one are known to exhibit an oscillatory character and produce a very high intensity noise. Such flows appear in equipment and installations, e.g. engines, valves, pipe systems and so on. Depending on the particular geometries the shock waves that appear interact with the existing boundary layers or free shear layers leading very often to flow instabilities. The different types of oscillations in internal transonic flows are systematically described in this paper. The mechanisms leading to the oscillatory self-excited flow character are elucidated and substantiated by experimental, and in some cases theoretical, data. As the flows are strongly geometry dependent, the following four most often encountered configurations are considered and their properties and peculiarities are pointed out: the curved duct, the convergent–divergent nozzle, the sudden enlargement of duct cross-section and the jet–cavity interaction. In the cases of the curved duct and the convergent–divergent nozzle the supersonic flow region is terminated by a shock wave which causes flow separation. At some values of the supplied to ambient pressure ratio, strong oscillations appear. During these oscillations strong acoustic disturbances (shock waves) are periodically transmitted upstream. Similarly, in a duct with a sudden enlargement of its cross-section, the downstream and upstream part of the flow are often affected by severe oscillations. In this flow geometry several mechanisms of instability have been detected. The jet and the upstream-facing cavity give a very wide variety of possible oscillatory configurations. Particularly in the transonic and supersonic flow regime, oscillations are very strong and lead to noise of high intensity and high gas temperatures at the base of the cavity.

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NOTATION

a—speed of sound
a_*—critical speed of sound
d, D—diameter
f—frequency
F, F_*—nozzle cross-section area
h, h_*—throat height
H, H_p—duct height
k—isentropic exponent
L_c—collar length
L, L_p—pipe length
M_1, M_2—flow Mach numbers ahead and behind the shock wave, respectively
p—pressure
p_1, p_2—pressure ahead and behind the shock wave, respectively
p_a—atmospheric pressure

p_e—exit pressure
p_o—supply pressure
p_w—base pressure
s—nozzle-resonator spacing
s_h—shock wave position downstream of the nozzle
Sh—Strouhal number
t—time
T—oscillation period
v_p—velocity of separation point
v_s—shock wave velocity
x, y—space coordinates
ρ—density
ρ_0—stagnation density
η—diffuser efficiency coefficient defined by equation 3.5

1. INTRODUCTION

Internal transonic flow is, in our definition, a compressible fluid flow in the vicinity of Mach-number one inside rigid boundaries. Among numerous technical applications such as valve or engine flow, this kind of flow (field) is well known for its oscillatory instabilities. Although the transonic flows around bodies or from nozzles into free space have similar flow instabilities, we have restricted our considerations in this paper, in the main, to confined flow fields.

Our definition of oscillatory instabilities is the following: for stationary geometrical boundary conditions, stationary solutions for the flow field are not observed. The flow is unsteady and oscillatory. To investigate this unsteadiness in experiments, suitable time-dependent control parameters are the pressure difference between the inside and outside of the duct, the static pressure measured by pressure transducers, and the density measured by high speed interferometric framing. The area ratio of the duct and other geometrical parameters, such as the length of the flow field in a characteristic direction, influence the observed frequencies because of the associated disturbance travel times.

Four typical flow configurations where the flows have a tendency to oscillatory behaviour, as defined above, are shown in Fig. 1. In the case of transonic flow conditions in these typical geometries, shock waves, boundary layer separation and free shear layers occur, which normally interact to give oscillatory instabilities. The limited space of the confined flow geometry supports these oscillations by several acoustic resonances of the upstream or downstream flow volumes.

In the case of curved ducts, the supersonic flow regime at the smallest cross-section and the largest curvature is usually terminated by a shock wave which tends to separate the boundary layer. This flow pattern influences the pressure downstream of the supersonic flow regime which is consequently increased. By this pressure change, the position of the shock wave and therefore the point of flow separation is displaced upstream.

In the case of flows in convergent–divergent channels, typically the configuration of a Laval nozzle or a supersonic diffuser, the interaction between the boundary layer and the shock is very similar. One difference from the previous case is the fact that the whole cross-section in the divergent part, upstream of the shock, is filled with a supersonic flow regime. In this nozzle flow, if the shock is oscillating with only small amplitudes, the upstream part of the nozzle flow is generally screened by the supersonic choking from the influence of the oscillations. Strong acoustic waves are radiated only downstream in this case. However, under some conditions, the strong disturbances (shock waves) can be periodically propagated upstream into the convergent part of the channel, choking the flow. This phenomenon is well known as 'buzz' in diffuser and nozzle flows.

In the nozzle, with a sudden enlargement of its cross-section, very often only the downstream part of the flow field is affected by the severe oscillations of the shock–shear-layer interaction which forms the main source of instability in this case. In this flow-geometry, several kinds of instability have been detected which are almost all based on shock–shear-layer interactions.

The last example of oscillatory instabilities in internal flow that will be considered is an interaction of a jet with an upstream-facing cavity. This example represents a very wide variety of possibilities for interaction of flow and cavities. Particularly in the transonic or supersonic flow regime these interactions are very strong and then lead to vibrations of large amplitude.

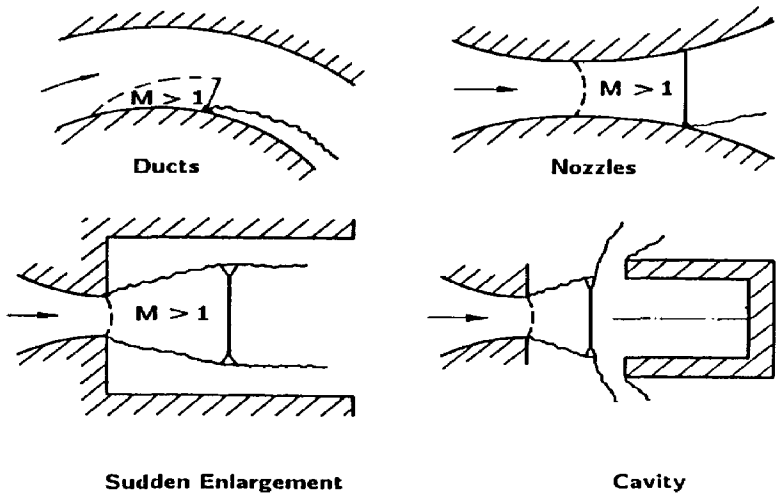


FIG. 1. Internal flow configurations.

The above mentioned phenomena have been intensively investigated in the last twenty years. The flow oscillation in the curved duct was studied experimentally only by Meier and Hiller.[37] In contrast, the Laval nozzle flow has been a subject of interest to many authors. The oscillating flow in the Laval nozzle has been studied experimentally by Meier,[38,39] Sajben *et al.*,[56–59] and Chen *et al.*,[12] and, theoretically, by Adamson *et al.*,[1,2] Chan and Adamson,[11] Culick and Rogers,[13] Liou *et al.*,[34,35] Messiter *et al.*,[43–45] Rein *et al.*[52,53] and Richley and Adamson.[54] Both the small disturbances method for the unsteady two-dimensional potential flow and the method of characteristics for the one-dimensional approach have been used to determine transient pressure traces and shock wave positions during the oscillations. Limits of the oscillations have been calculated based on the pressure distribution for one-dimensional stationary flow.[39]

The oscillating flow in the duct with an abrupt increase of internal cross-section has been studied by many authors both in subsonic and supersonic flow regime. In the first instance, the jet flowing out of a duct of circular cross-section was the subject of interest. It has been observed[16–19] that the mixing rates of the jet with the surrounding air increase considerably due to the high turbulence of the oscillating flow. This produces a faster decay of the mean velocity along the jet and larger sound pressure fluctuation intensities in the near field. The limits of the oscillations depend on the supply to ambient pressure ratio and the dimensions of the system. These limits have been determined experimentally.[62,69] Three oscillation types have been distinguished and explained. Two of them are symmetrical and the third asymmetrical. In the supersonic case, most experiments have been performed using a duct of rectangular cross-section.[3–7,40,41,68] This type of duct enables flow visualisation to be made. Oscillations in supersonic flow are accompanied by moving strong shock waves and separation regions. The oscillation modes in this case differ from one another in the shock configurations and their intensity.

The acoustical behaviour of jet-cavity interaction was discovered by Hartmann[15] and the thermal effects by Sprenger.[67] Some other authors[24–26,28–32,49–51,61,65] who have explored this problem have considered the influence of many flow and geometrical properties of the system. In many cases the oscillation mechanism is controlled additionally by flow–acoustic interaction. The usual theoretical approach[26,46,70] is based on a one-dimensional model of the flow in the cavity. The influence of the boundary conditions at the open end of the cavity in these models is crucial. These conditions depend on the flow properties in the free space between the nozzle and the cavity. Because this flow is unsteady and three-dimensional it is very difficult to determine the proper boundary conditions at the cavity inlet. It has been achieved only for low frequency modes hitherto.

In the following we give some detailed descriptions of the phenomena for the above geometries.

2. CURVED DUCT

The unsteady flow in a curved duct having a supersonic region embedded in the subsonic flow field, shows great similarity to certain forms of buffeting on wings and blades. This problem was investigated experimentally in a flow which, under certain conditions, was steady with shock-free recompression[33] by Meier and Hiller.[37] The channel boundaries in their experiments were streamlines of an exact solution of the basic gasdynamic equations. This solution describes the compressible flow around a half plane and has been obtained by Ringleb[55] using a Molenbroek–Tschaplygin transformation. Some of the streamlines of the exact solution are shown in Fig. 2. The chosen boundary streamlines are emphasised. The supersonic region fills one quarter of the smallest channel cross-section. The largest velocity occurs at the vertex of the convex boundary ($M = 1.25$); Fig. 3 shows a schematic drawing of the channel.

The channel flow was drawn into a vacuum chamber and was fed with dried atmospheric air. The flow volume was controlled by a variable nozzle in which the flow was accelerated to the velocity of sound. The boundary layer near the vertex of the convex boundary was controlled by suction. In this section, the duct wall consisted of polished sintered sheet metal.

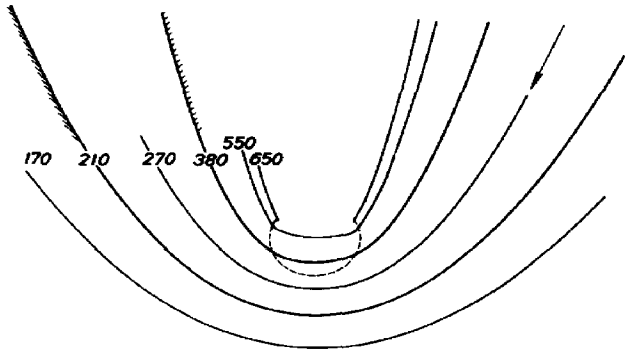


FIG. 2. Stream lines of the transonic potential flow; numbers indicate the maximum velocity on the streamline in m/s. (– – –) sonic line (from Ref. 39).

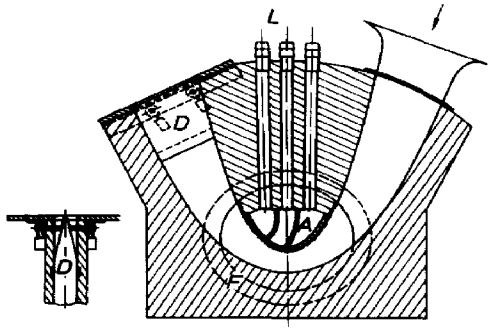


FIG. 3. Curved duct, A—suction, D—control nozzle, F—glass window, L—pipes for suction (from Ref. 39).

The comparison between an interference mapping obtained from the exact solution and a photograph of the flow field under optimal conditions shows the degree of agreement between the theoretical flow field and its experimental realisation (Fig. 4). If, starting from this steady shock-free flow, the velocity is increased, weak shocks will be generated at the end of the supersonic region (Fig. 5). As may be seen from the photographs these shocks are unstable in location and strength.

Further increase of the Mach number leads to stronger shocks (Fig. 6). The pressure jump at the root of the shock becomes so large as to cause the boundary layer to separate (frames 1 and 2). The separation bubble behind the shock rapidly increases. Then the higher pressure in the downstream part of the duct causes the back flow in the separation bubble to rise and thus pressure increases just behind the shock. In consequence the shock wave moves upstream. The point of separation of the boundary layer will move upstream together with the shock (frames 2–4). The shock strength decreases during this phase and the shock reaches areas with small pressure gradients (frame 5). The flow therefore becomes attached to the wall again. The dead water bubble moves downstream (frame 6). The velocity of the flow field now has a minimum. Thus the shock is eliminated completely. In the following acceleration phase the supersonic region grows again (frames 7 and 8). As

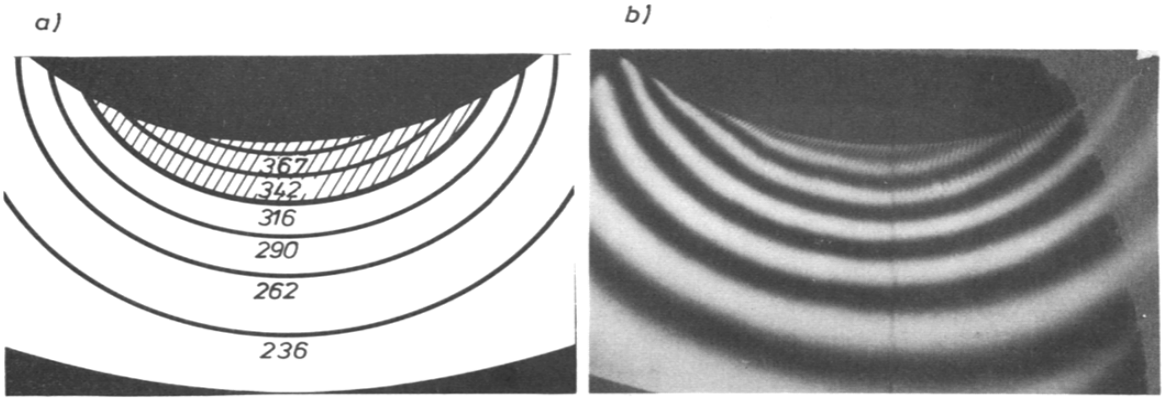


FIG. 4. Calculated (a) and observed (b) interference fringes; numbers indicate velocity in m/s. Flow direction is from right to left. Mach number in the vertex of boundary $M = 1.25$ (from Ref. 39).

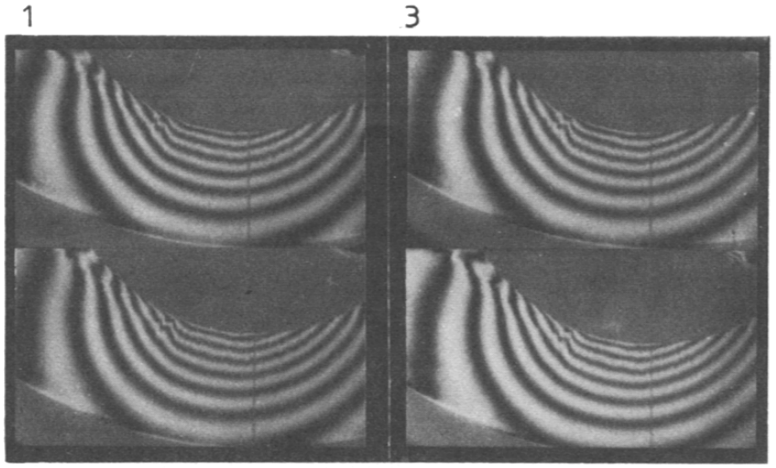


FIG. 5. Weak unsteady shocks at the end of the supersonic region. Flow direction is from right to left. Interframe time $\Delta t = 0.4$ ms (from Ref. 38).

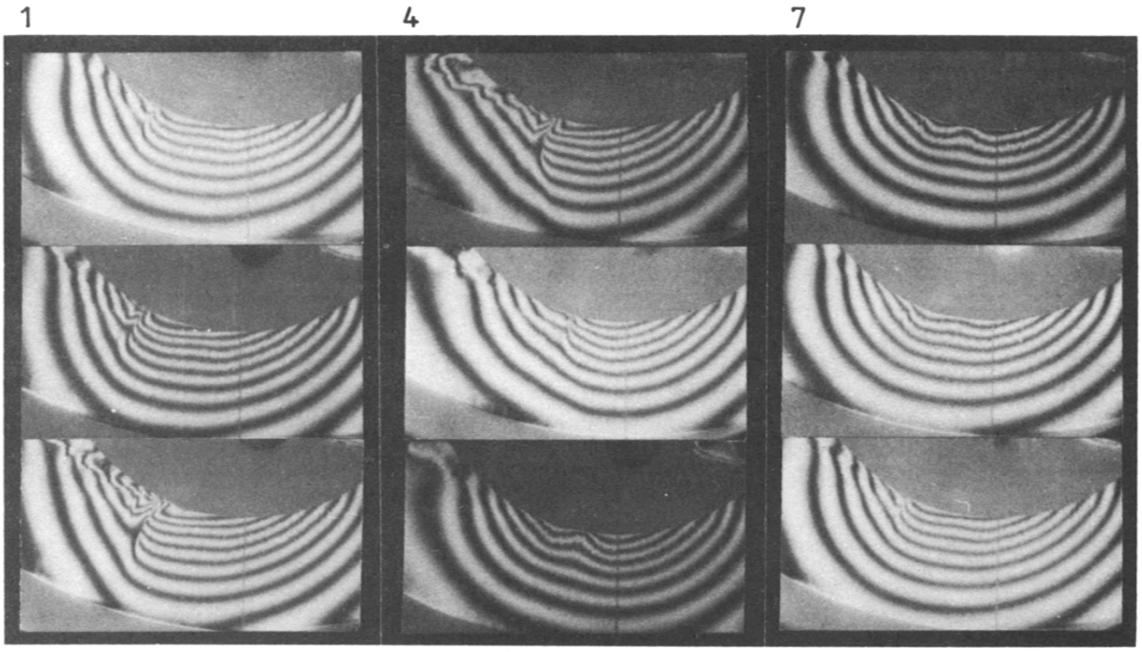


FIG. 6. Full cycle of an oscillation of unsteady transonic flow. Flow direction is from right to left. Interframe time $\Delta t = 0.7$ ms (from Ref. 38).

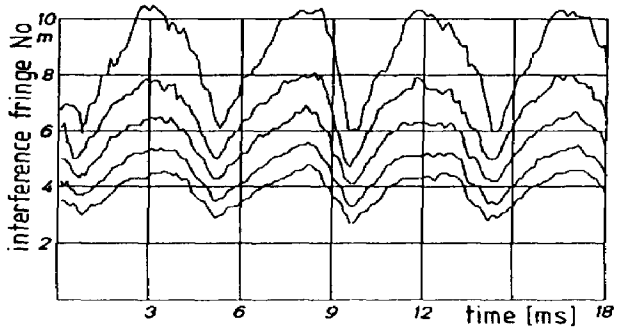


FIG. 7. Relative density traces in five scanning points; here density is measured by order of interference fringes (from Ref. 39).

soon as a certain Mach number at the end of the supersonic region is exceeded, a new shock forms (frame 9). The whole cycle as described in this example was repeated with a frequency of 180 Hz.

The essential element of the instability and vibration generation is the separation of the boundary layer coupled to the shock. To support this argument the wave propagation in the channel has been investigated. The time dependence of the density has been inferred from the interferograms for approximately fifty scanning points. An example of those time-traces is given in Fig. 7.

By cross-correlating these time sequences, phase relationships for the density waves in the flow may be obtained, which lead to a so-called phase plan. Figure 8 shows the phase plan corresponding to the unsteady flow shown in Fig. 6. This indicates that the wave center is in that part of the flow, where the boundary layer is induced to separate by the shock.

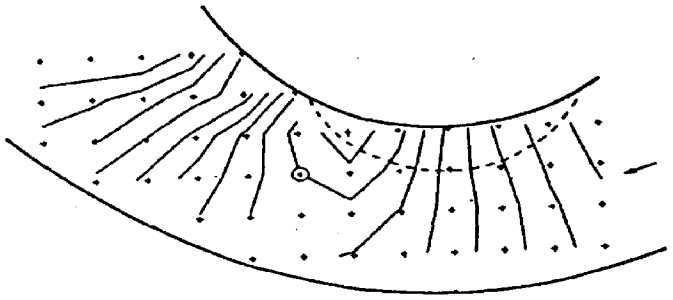


FIG. 8. Phase plan (from Ref. 39).

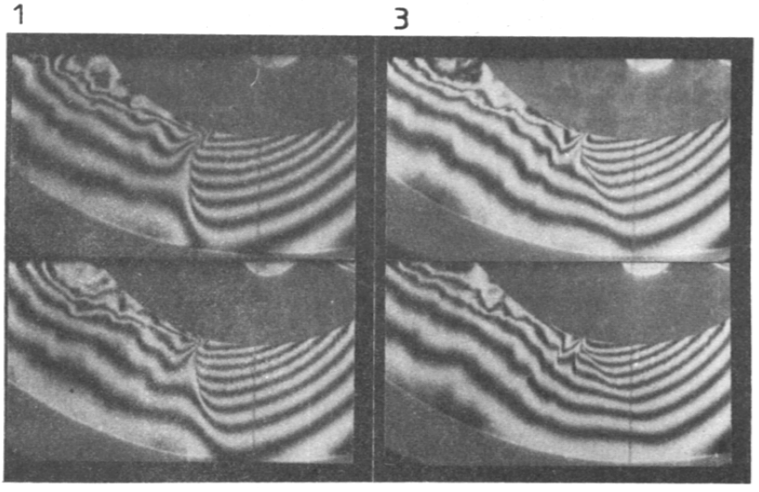


FIG. 9. Splitting of a shock wave. Flow direction is from right to left. Interframe time $\Delta t = 0.1$ ms.

At very large flow volumes strong shocks may be observed, which penetrate deeply into the subsonic region while moving against the flow (Fig. 9). As soon as the root of the shock comes to rest (Fig. 9, frame 3) the shock is split into two (frame 4). The unsteady part moving further against the flow leaves an almost steady normal shock downstream, as shown in frames 1 and 2. The fact that the shock separates into two parts may be explained by the difference of their respective components of speed of propagation in the direction of the flow and by the difference of flow speed along their paths.

Apart from the splitting process however, the propagation of the shocks is in accordance with a geometric acoustic model for a high wave number approximation. Figure 10 shows the calculated propagation of a perturbation situated initially at the vertex of the known flow field. It can be seen that the perturbation initially moves downstream in the supersonic region, until it takes on the Mach angle, and may propagate into the supersonic region. It must not be concluded, however, that energy may be transported upstream into the supersonic region. Figure 11 shows the approach for a different initial position of the perturbation in addition to the orthogonal trajectories, giving the direction of propagation of the wave, which gives the direction of the group velocity. This makes clear that energy propagation has a strong component in the direction of the shock wave and is always directed downstream into the supersonic flow regime. This, at the same time, explains the rapid decrease in the strength of the shock during the propagation (Fig. 9).

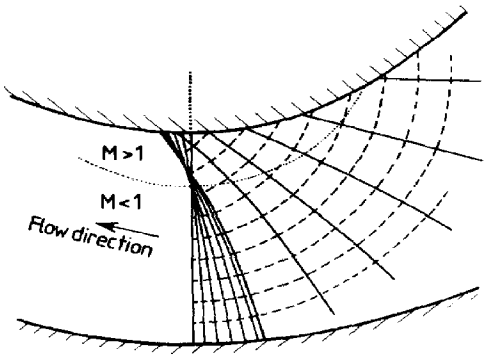


FIG. 10. Calculated motion of a shock wave. Initial position of the shock is the vertex of the channel. (——) momentary shock position, (– – –) phase velocity direction, (. . .) sonic line (from Ref. 39).

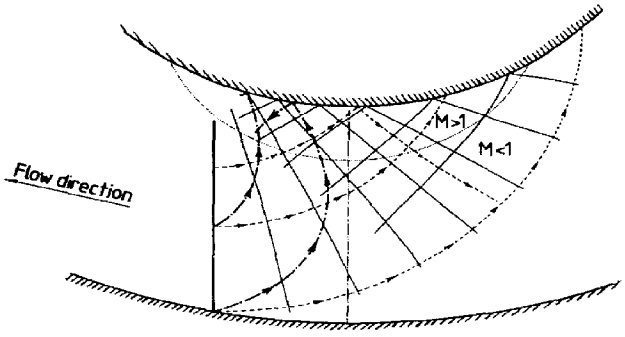


FIG 11. Calculated motion of a shock wave. (——) momentary shock position (also reflected shock), (– – –) phase velocity direction, (–·–·) group velocity direction, (. . .) sonic line (from Ref. 39).

3. CONVERGENT–DIVERGENT DUCT

The performance of supersonic aircraft propulsion systems is limited by the occurrence of self-excited, unsteady fluctuations in air inlets at off-design operational conditions. These fluctuations are well known as strong dynamic distortions and buzz. The flow in wind tunnels exhibits similar features.

The properties of the fluctuations were explored by investigating transonic flows in convergent–divergent channels under moderate inlet to outlet pressure ratios corresponding to weak shocks near the throat.

3.1. EXPERIMENTAL OBSERVATIONS

As is mentioned in the introduction, there are only a few experimental investigations[12,38,39,56–59] and these did not examine all of the possible events of the flow field under consideration.

In Meier's experiments[38,39] a two-dimensional nozzle 100 mm wide (Fig. 12) has been used. The divergent part of the side walls of the nozzle had a hyperbolic type contour $y = 0.5\,h_* (1 + x^2/b^2)^{0.5}$, where h_* (50 mm) is the throat height, and $b = 93$ mm.

Dried air was sucked through the nozzle from a plastic bag reservoir. Just after the nozzle was a receiver chamber, in which the pressure was maintained constant by operation of a control valve. The flow in Meier's[38,39] experiment was always separated due to the positive pressure gradient, either through the shock wave, or along the divergent part of the

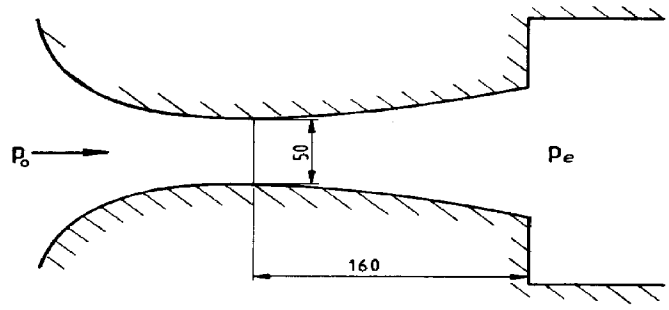


FIG. 12. Nozzle model; spanwise dimension of channel—100 mm (from Ref. 38).

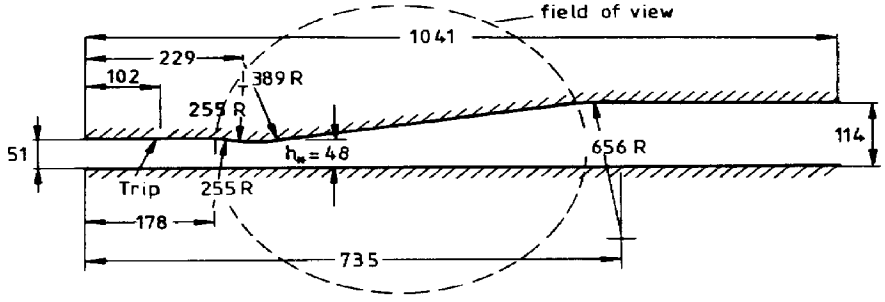


FIG. 13. Diffuser model; spanwise dimension of channel—152 mm (from Ref. 12).

nozzle downstream of the shock. In both cases, the separation bubble extended to the receiver chamber. Hence the flow downstream of the shock wave was strongly influenced by the receiver pressure due to the mass momentum and energy transport, throughout the separation bubble.

In contrast to the nozzle flow investigated by Meier, the divergent duct used by Sajben, Kroutil, Bogar, and Chen[12,56–59] (see Fig. 13) was extended with a relatively long constant cross-section part. This configuration enabled the flow to reattach downstream, causing the separation bubble to be closed. In this case the receiver pressure influences the upstream flow only through acoustic waves.

The effect of the receiver pressure is crucial for the behaviour of unsteady flow in the divergent duct. This is the reason why Meier's experiments cannot be directly compared with those of Sajben *et al.* Generally, the experimental investigation made by Meier can be taken to model the transonic flow in the Laval nozzles, whereas Sajben *et al.*'s investigations are closely connected to diffuser flow. Here, we will display briefly the most significant results of both sets of experiments.

Figure 14 shows interferograms of the observed oscillation in the Laval nozzle (see also Fig. 12). The pressure ratio is $p_e/p_o = 0.76$, where p_e and p_o are the rest pressures in the receiver and supply reservoirs. Frame 1 in Fig. 14 shows a shock wave just at the position of the smallest cross section, which is marked by the small vertical line. This shock later penetrates the subsonic intake flow, leaving the intake of the duct as an acoustically loud ('sonic bang') wave. Frame 2 shows a completely subsonic flow in the nozzle. In this case of subsonic flow, separation is maintained by relatively small pressure or density gradients (see frame 2). The boundary layer is separated only at one boundary of the duct. This is due to another type of instability, which, in the case of two separation bubbles on opposite sides, causes the larger one to grow due to centrifugal pressure differences across the curved

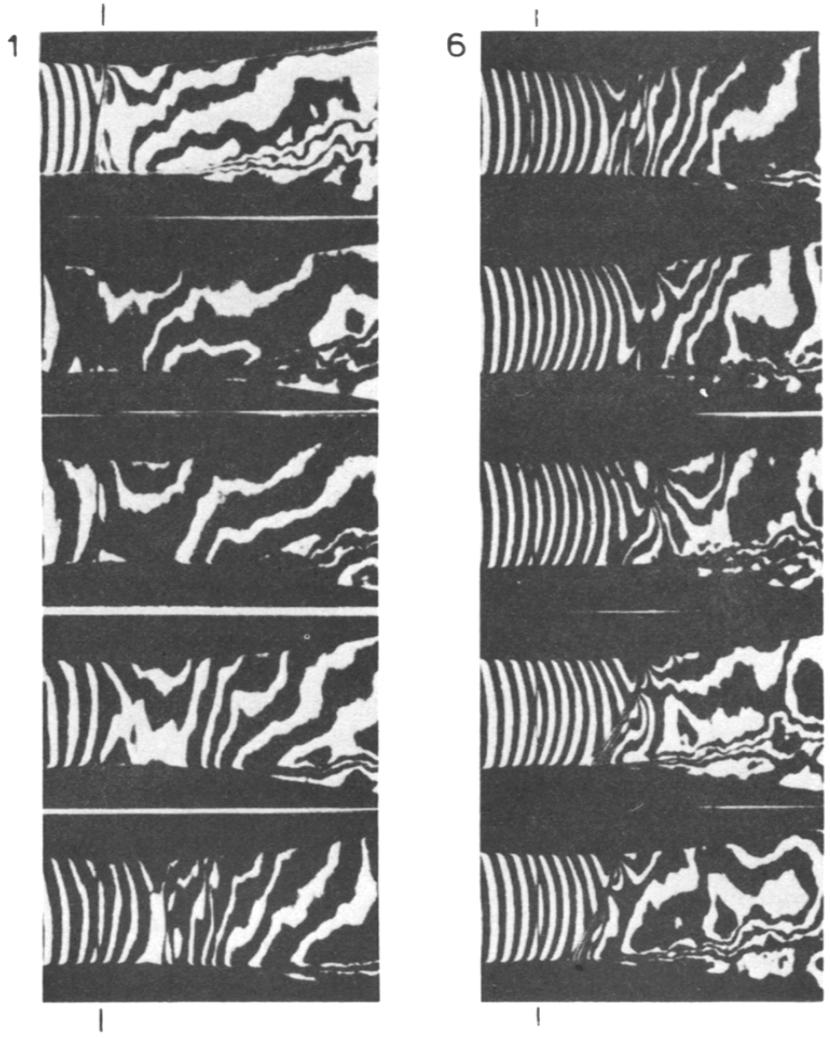


FIG. 14. Interferograms of oscillating flow in a Laval nozzle, $h_* = 50$ mm. Pressure ratio $p_0/p_1 = 1.32$. Interframe time $\Delta t = 0.5$ m/s (from Ref. 38).

stream lines, and the smaller one to decrease. Thus, at an early stage, a random perturbation may determine on which side boundary layer separation takes place. By means of the applied pressure ratio the flow is accelerated in the frames 3–5. In this acceleration phase the boundary layer is stabilised and the shock wave is not concentrated in a fixed position (see frame 5). If the shock is formed (see frame 7), the separation point switches to the position of the shock root. Thus the shock gets additional back pressure and the separation bubble starts growing (see frames 7–9). The shock wave then penetrates the whole supersonic flow, transforming it to subsonic flow (see frames 8–10 and 1). So the cycle is completed.

The density fluctuations, shown in Fig. 15a, are evaluated from interferograms. The very great changes in trace 2 of up to 0.35 in ρ/ρ_0 are due to the fact that the flow Mach number changes from $M = 1.2$ in the supersonic flow phase to $M = 0.7$ in the subsonic flow phase. This indicates the one-dimensional behaviour of this duct flow with no further possibilities of radiation or equalisation of time-dependent distortions other than in the axial direction. In other words, although the behaviour of the flow is a reaction to a separation at only one of four boundaries, the reaction influences a single cross section nearly uniformly. This is the main assumption for the gasdynamic descriptions of the mechanism.

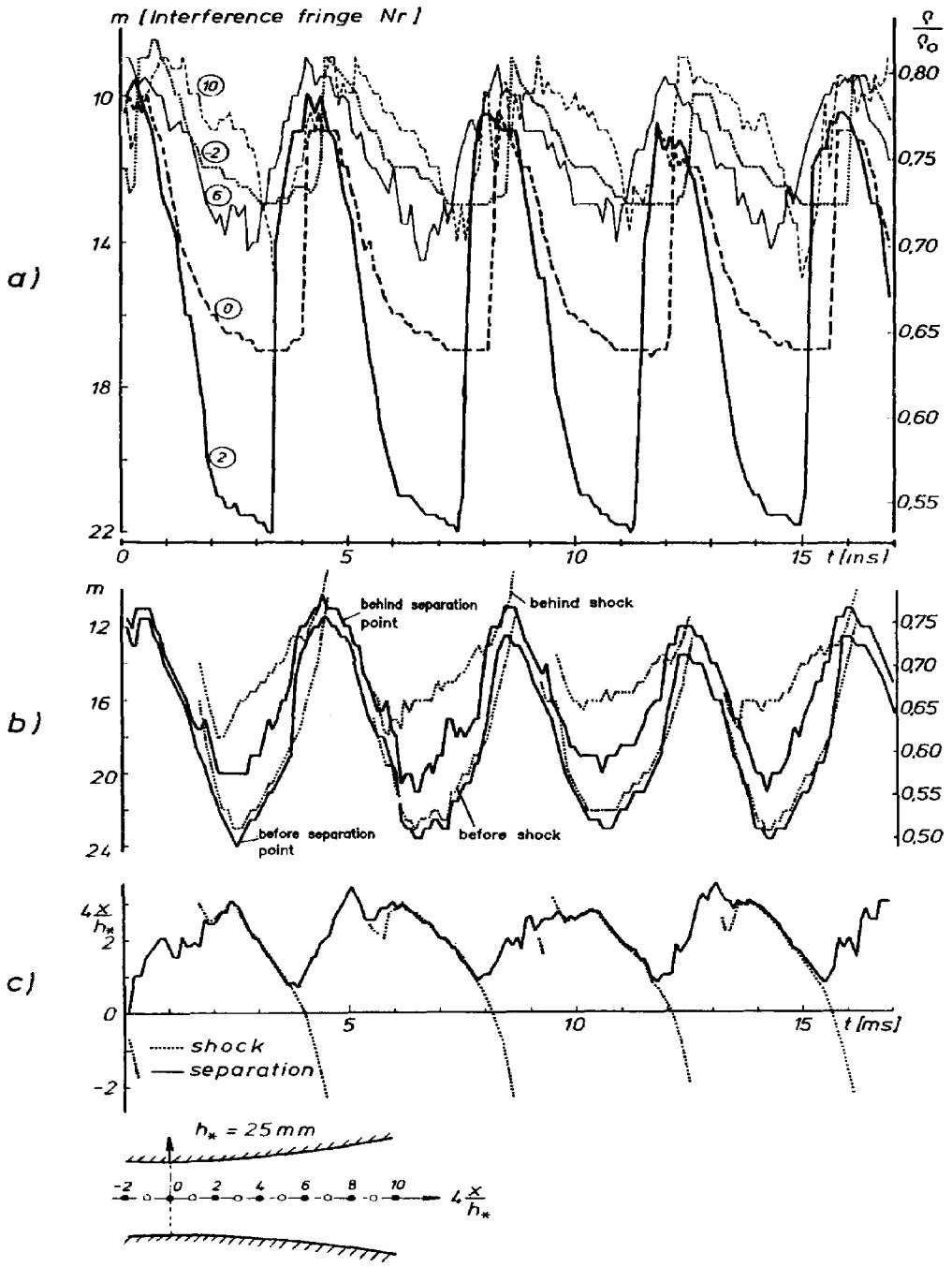


FIG. 15. Time dependence of the density at five points (a), the density before and after shock or separation point respectively (b), and position of shock waves (c), (from Ref. 39).

A plot of the densities just before and behind the shock wave and separation point is shown in Fig. 15b. Differences between these values become great at the end of the acceleration phase. The differences become smaller during the deceleration phase. It is interesting to note that the density itself behind the shock wave is still growing during the

deceleration. This is due to an amplification of the driving upstream pressure wave. In Fig. 15c is a plot of the positions of the shock and the boundary layer separation point versus time. It is obvious that the downstream velocity of the separation point (v_p) in the acceleration period is fairly high. Typical values of velocity $v_p = 30$ m/s are observed. These values correspond to velocities in the attached boundary layer. The outer shape of the separation bubble is changed with a much higher velocity. This corresponds to the convection velocity of turbulence elements, and typical values of $v_p = 100$ m/s are observed from interferograms. The motion of the shock wave, also plotted in Fig. 15c, is coupled to the motion of the separation point in the deceleration phase up to a certain position downstream of the smallest cross-section. Here they split with remarkable speed. This sudden splitting is associated with the tendency of the separation point to move to a more downstream position where the duct geometry and boundary layer characteristics are more favourable to separation, i.e. the coupling between the shock wave and the separation point becomes too weak to carry the separation further upstream. After the splitting, the shock wave increases its velocity from typical values of $v_s = 25$ m/s up to $v_s = 50$ m/s, penetrating the first part of the subsonic inlet flow.

The time sequences of density changes of the nozzle flow are plotted in Fig. 16. The motion of the shock wave and the subsonic flow regime as a whole can clearly be recognised

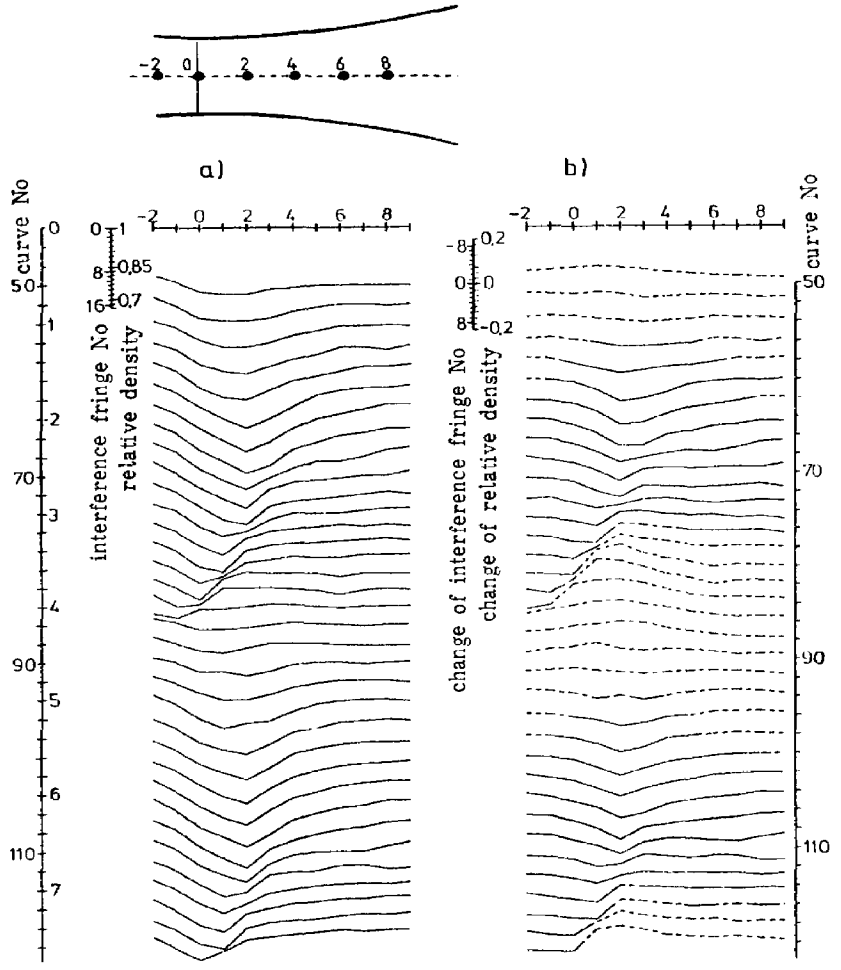


FIG. 16. Time sequence of density (a) and density changes (b) at the centre-line of the Laval nozzle. High density in the changes is marked by dashed lines (from Ref. 39).

from the left-hand density sequence. In the right hand sequence one can see that the density increase starts just behind the shock wave and is propagated up- and downstream (note that density is greater than the time average indicated by dashed lines). The acceleration (density decrease) starts in the same area of the duct and is propagated in the upstream direction; deceleration shows the opposite behaviour. All of these phase velocities are in the interval $40 \text{ m/s} < v_p < 100 \text{ m/s}$.

One can get some further ideas about the oscillation mechanism from a one-dimensional model of the Laval nozzle flow. Figure 17 shows the pressure distributions for nozzle flow without and with separation, plotted as continuous and dashed lines, respectively. One can get additional information about the mechanism from this diagram by introducing some knowledge of the boundary layer separation and pressure conditions. Curve A shows a pressure distribution in the duct corresponding to frame 2 in Fig. 14, with subsonic flow and separation at the marked point. If we now consider the oscillation mechanism in a quasistationary model, we see that one can reach distributions like B and C (Fig. 17) without much change in the inlet and outlet pressure. It is important to note that the shock waves B and C satisfy the pressure relation given by:

$$\frac{p_2}{p_o} = \frac{2kM_1^2 - (k-1)}{k+1}\left(\frac{2}{2+(k-1)M_1^2}\right)^{\frac{k}{k-1}}. \tag{3.1}$$

This pressure is plotted as a dotted line for different front Mach numbers M_1, as a function of the dimensionless length l/D or cross-section F/F_* in Fig. 17

$$\frac{F}{F_*} = \frac{1}{M_1}\left[\frac{2}{k+1}\left(1+\frac{k-1}{2}M_1^2\right)\right]^{\frac{k+1}{2(k-1)}}. \tag{3.2}$$

The value of $k = 1.4$ for air leads to a maximum of the back pressure curve at $M_m = 1.48$. We see from the diagram that there is a difference between the back pressure of the shock waves and the pressure distribution of the separation bubble, which can be overcome by a very steep increase of pressure along the corresponding dashed curve. This steep increase after the pressure jump of the shock wave will not be tolerated by a normal boundary layer in a divergent duct. Thus the separation point jumps to the shock root. This has two consequences with respect to the pressure distribution. The pressure gradient on the separated flow is increased; so the back flow is accelerated and the bubble grows. The pressure jump in the shock wave becomes too high for a stationary shock and it moves upstream, taking the separation point and, therefore, its instability with it. This situation is

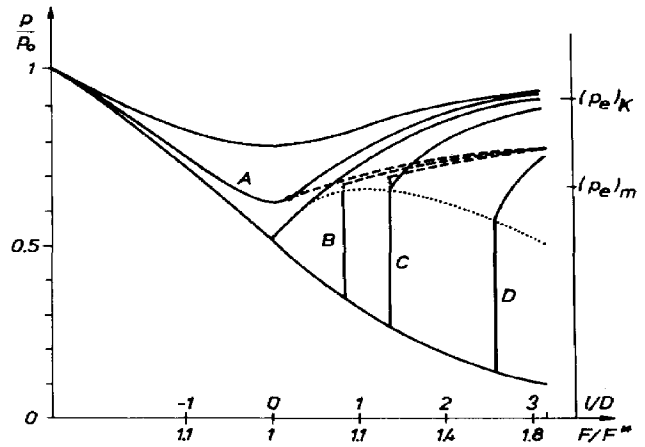


FIG. 17. Pressure distributions in Laval nozzle for steady flow case versus dimensionless length or cross-section area (from Ref. 38).

indicated in Fig. 17 by extension of the shock wave to the point just above the dotted line. From these considerations we also obtain some information about the instability regime of transonic flows for oscillations described here. If we take, for instance, a receiver pressure p_e, lower than the maximum $(p_e)_m$ of the back pressure curve, leading to a pressure distribution like curve D, no instability will occur. With the pressure below $(p_e)_m$, if the shock wave is moved upstream, the pressure gradient along the separated region tends to zero and the back flow decreases. It causes the shock wave to move downstream again. Another limit for oscillations is obviously the critical pressure $(p_e)_K$. The nozzle flow can reach supersonic velocity, which is essential for the mechanism, only with a discharge pressure below $(p_e)_K$. In fact this limit is shifted down to a certain value lower than $(p_e)_K$ by pressure loss in the separation, which depends on the apparatus. From this knowledge one can deduce an essential criterion of instability for this type of oscillation.

In terms of Mach number

$$1 \leq M_1 \leq \left(\frac{k+3}{2}\right)^{0.5} = 1.483 \quad (\text{for } k = 1.4). \tag{3.3}$$

In terms of pressure

$$\frac{(p_e)_K}{p_o} \geq \frac{p_e}{p_o} \geq 2\left(\frac{2}{k+1}\right)^{\frac{k+1}{k-1}} = 0.6698 \quad (\text{for } k = 1.4). \tag{3.4}$$

Both the limits are in excellent agreement with experimental results for the Laval nozzle flow. The limit for penetration of shock waves in the subsonic intake flow was found for $p_e/p_o = 0.66$. The disappearance of oscillations was found for $p_e/p_o = 0.86$. The value for $(p_e/p_o)_K = 0.925$ for this nozzle is greater: the difference is due to the pressure loss by separation. This effect can be expressed by diffuser efficiency coefficient η for the subsonic flow downstream of the normal shock wave. The coefficient η is defined here by the relation

$$\frac{p_e}{p_2} = \left(1 + \eta \frac{k-1}{2} M_2^2\right)^{\frac{k}{k-1}} \tag{3.5}$$

where suffix 2 denotes conditions behind the shock wave. $\eta = 0$ for fully separated and $\eta = 1$ for unseparated isentropic flow behind the shock.

Matsuo *et al.*[36] extended the criterion of Eq. (3.3) taking into account both M_1 and η (Fig. 18).

The phenomenon, typical in Meier's experiments with Laval nozzles, when shocks propagate upstream, cross the throat and eventually disappear in the convergent section, was only occasionally observed by Sajben *et al.*[56,57] in their diffuser flow. Four typical flow patterns for transonic flow which were identified by these authors are displayed in Fig. 19.

Mode (i), $M_1 < 1.1$, is an intermittently shock-free, multiple-shocklet pattern, but in this case shocks never cross the throat. Presumably, a sonic line and a small supersonic zone of

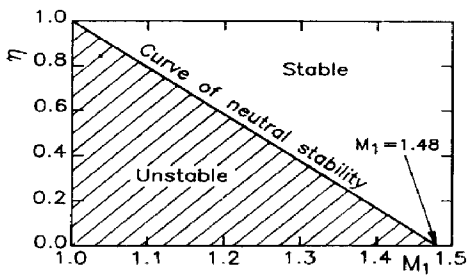


FIG. 18. Nomograph of stability of Laval nozzle flow (from Ref. 36).

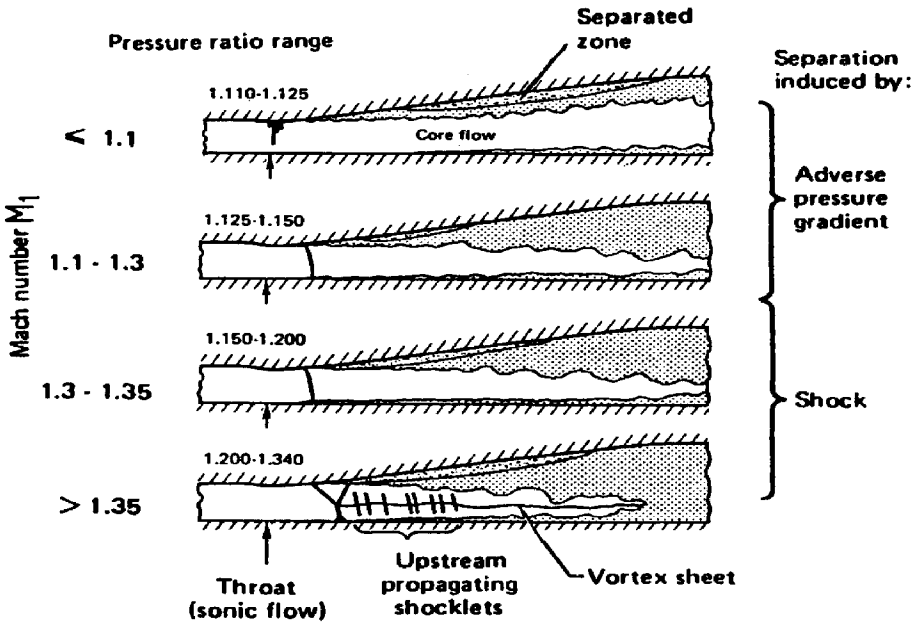


FIG. 19. Typical flow patterns in diffuser (from Ref. 59).

varying size exist at all times. The supersonic Mach numbers are close to one, and no visible shocks occur for some fraction of time. For these reasons mode (i) is characterised as persistently transonic, intermittently shock-free flow.

Mode (ii), $1.1 < M_1 < 1.3$, displays a single, approximately normal shock at all times. Multiple, weak shock patterns are absent. The shock is somewhat curved, and the shock angles close to the wall are large enough for the shock to be described as oblique, with a pressure rise considerably below that for a normal shock. The shock does not cause immediate separation.

Mode (iii), $1.3 < M_1 < 1.35$, appears where the separation point abruptly moves upstream and assumes the same position as the shock foot while the length of the separation bubble is nearly doubled.

When the flow Mach-number M_1 increases over 1.35, the top of the shock intermittently splits into a large lambda pattern, with the separation point located near the intersection of the upstream leg with the wall—mode (iv). The vortex sheet originating at the bifurcation point is clearly visible and persists without appreciable diffusion throughout the core flow. The fraction of time spent in the lambda formation gradually increases until a full-time lambda pattern exists. The length of the bubble grows with M_1 throughout this range.

The growth rates of top and bottom boundary layers increase gradually with pressure ratio until shocks appear. The presence of shocks is accompanied by rapidly thickening boundary layers. In modes (ii) to (iv) the top and bottom layers merge within 4 to 6 throat heights from the shock. Shock wave changes with increasing M_1 are accompanied by changes of length and location of the separation bubble. For low M_1 the separation is induced by the pressure gradient. The bubble is long initially, and its length remains constant until shocks appear at $M_1 = 1.1$, where separation is still caused by the adverse pressure gradient and the separation point is well downstream of the shock. Coincident with reaching supercriticality, the bubble begins to shrink as the pressure ratio increases and the shock becomes stronger. This gradual trend persists to $M_1 = 1.3$ when the separation point moves upstream to coincide with the location of the shock foot. Simultaneously, the reattachment point is displaced downstream, thereby nearly doubling the length

of the bubble. The pressure gradient-induced separation is thus replaced by shock-induced separation.

It is evident that the separation bubble, and thus the fluctuation, depends on the boundary layer thickness approaching the diffuser throat. To check this dependence Sajben *et al.*[59] have made experiments with the diffuser described above using a sequence of four inlet boundary layer thicknesses having the same shape factor. They have concluded that boundary layer thickness has a significant but moderate influence on both the time and the fluctuation flow properties when flow separation occurs well downstream of the shock, i.e. separation induced by the positive pressure gradient in the subsonic flow. As the boundary layer increases, separation occurs earlier, reattachment occurs later, static and total pressure recoveries are reduced, and the three-dimensional features of the flow in the duct are intensified. Shock displacement amplitudes increase dramatically, but frequencies increase slightly. Surface pressure fluctuations change minimally.

Shock Mach numbers greater than 1.3 involve a pressure jump large enough to cause immediate separation. An increase of boundary layer thickness under this condition generally results in consequences similar to those for weak shocks.

To have more insight into the mechanisms of diffuser flow oscillations, Chen *et al.*[12] have made some space–time correlation maps of pressure signals from transducers distributed on the bottom wall (see Fig. 13). An example of such a map for the shock separation case is shown in Fig. 20, where the time (t) coordinate is normalised by the peak frequency (f_{max}) of the reference signal (from the transducer located closest to the time-mean shock position). Contours in the figure display constant values of the correlation coefficient obtained by cross-correlation of the reference signal with itself (correlation coefficient equal to one) and with the remaining pressure signals. If the contour map is thought of as a three-dimensional surface, then the locus defines a ridge-shaped topological feature and will be referred to as a ridge in this paper. The inverse of the ridge slope on the x, t plane is commonly interpreted in experimental practice as the propagation (or convection) speed of the perturbations.

Within the oscillating range of the shock (left part in Fig. 20), where the pressure signals are dominated by the shock passage over the transducers, all correlation contours and thus the ridge show a negative slope, i.e. an upstream motion of the disturbances. Outside the shock oscillation range, the ridge has positive slopes, indicating that the downstream motion of the perturbations dominates in this case. The speed suggested by the slope of the shaded areas ranges from 30–70 m/s; in all cases it is well below the core flow velocity and negligible compared with the speed of downstream propagating pressure waves. The only reasonable explanation is that the perturbations are convected within the boundary layer, largely on the top wall.

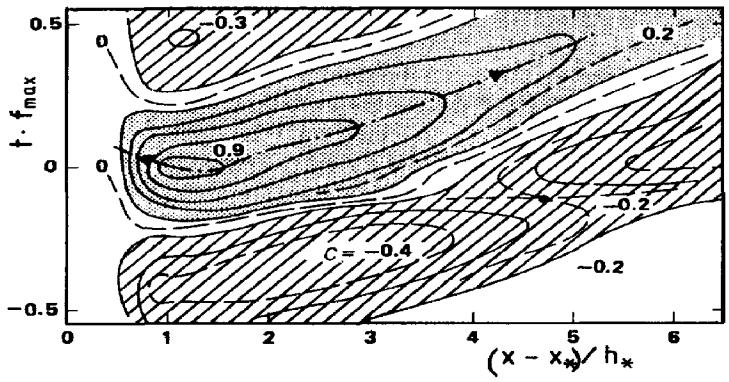


FIG. 20. Contour map of space–time correlation coefficients (C) for pressure ratio $p_o/p_a = 1.15$. Time delay scale normalised by frequency of reference signal $f_{max} = 70$ Hz. Increment of C between adjacent solid lines is 0.2; intermediate values are shown as dashed lines. Shaded and cross-hatched areas emphasize positive and negative values of C, respectively (from Ref. 12).

Two types of pressure waves described above were also observed in Meier's experiment[38] (Fig. 16); however, the downstream propagating waves were, in this case, not as distinct as the upstream ones displaying the shock motion.

It should be noted that the investigations of Sajben *et al.*[56–59] concerned the stable range of shock wave position for Laval nozzle flow ($1 < M_1 < 1.48$ for $k = 1.4$). The oscillations which were observed in this range for diffuser flow are relatively small and appear due to flow reattachment, downstream of the shock wave (closed separation bubble). As mentioned above this was because of the flow in the constant cross-section channel extending the diffuser. Flow oscillations in the diffuser extended by a constant cross-section part show close similarity to the so-called base pressure oscillations in a duct having an abrupt expansion of cross-section. This case is described in Section 4.

3.2. Theoretical Modelling

Kantrowitz[27] was the first to prove that smooth deceleration of the flow through the speed of sound is impossible in a convergent–divergent duct and occurs through a shock wave. He stated that the shock wave is unstable in the convergent duct and stable in a divergent one. However, the simple one-dimensional analysis of transonic Laval nozzle flow leads to the conclusion[38,39] that the shock wave is also unstable in the divergent part in cases when it appears sufficiently near the throat (see Eq. (3.3)); otherwise the instability can be induced only by external excitation. These cases were considered theoretically under various assumptions.

Shock wave positions for the case of self-excited oscillations were calculated by Rein *et al.*[53] using the method of characteristics for one-dimensional flow of an inviscid gas. Shock induced separation was assumed for $M_1 > 1.3$ which is in agreement with experimental[58] (Fig. 19) and theoretical investigations[8] for turbulent boundary layer shock interactions. According to this assumption the separation splits from the shock and moves downstream at $M_1 = 1.3$. For the velocity v_p of the downstream motion of the separation point, the following formula was assumed:

$$v_p = 0.5\, v_a \exp(-\beta x_a) \tag{3.6}$$

where x_a and v_a are the loci of the separation point and the flow velocity in the basic flow field respectively. The decay constant β is a free parameter. This formula is based on the assumption that the separation bubble is moving downstream with about half of the flow velocity if the shock wave has disappeared in the upstream direction. Because the receiver pressure in this situation is too small for nonseparated subsonic flow, the separation point is not able to move downstream, up to the end of the nozzle. The slow approach of the point of separation to a fixed final position close to the end of the nozzle is modelled by an exponential decay of the velocity with increasing x.

The calculated shock and separation point paths for the nozzle used by Meier[38,39] (see Section 3.1) are shown in Fig. 21. The value of the coefficient β was chosen to fit the calculated and measured oscillation frequency. In this case $\beta = 0.5$. The calculated pressure traces at three cross-sections of the nozzle are shown in Fig. 22.

They can be qualitatively compared with curves (-2, 0, 2) in Fig. 15a representing the density fluctuations. Such a comparison between pressure and density is possible because the changes of entropy are comparatively small. In both cases pressure and density fall very quickly to minimal values and then stay constant over a relatively long time. The reason for this is the growing supersonic flow regime which is choking the nozzle and preventing any propagation of disturbances from the end of the nozzle to upstream positions. The calculated values of minimum pressure and density agree very well with the measured values ($\pm$ 3%). For the maxima the error is a little bit larger ($\pm$ 10%), because in the calculations the assumption was made that the pressure gradient between the point of separation and the end of the nozzle is zero, which is not true in the experiment.

Although the method of Rein *et al.*[52] requires the value of β to be established a priori or from experiment, it can be useful in predicting the distribution of oscillation amplitudes.

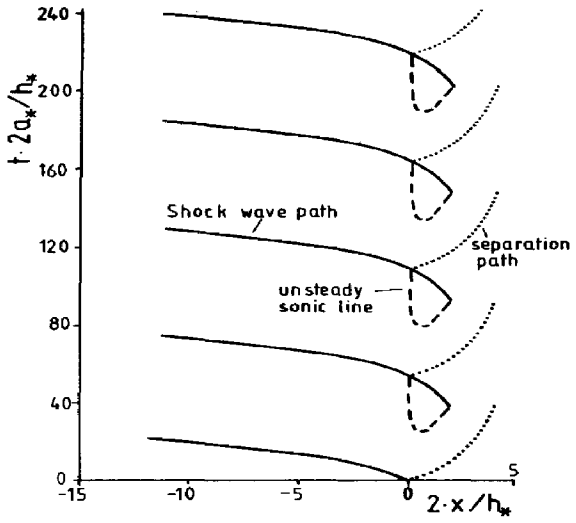


FIG. 21. Shock and separation point paths (from Ref. 52).

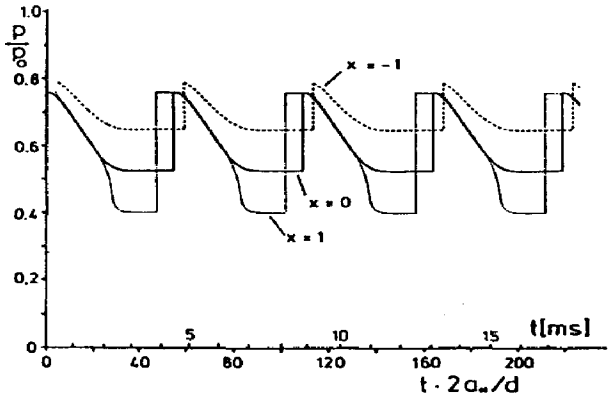


FIG. 22. Calculated pressure traces at the three cross-sections along the nozzle—see Fig. 15 (from Ref. 52).

More attention in the theoretical approach has been focused on the case of external excitation of the downstream flow. These types of excitation may be caused by combustion instabilities, oscillating blade wakes, etc. Because this paper is concerned in the main with the self-excited oscillations, we will review this case only briefly.

Culick and Rogers[13] treated small pressure disturbances in a first approximation as acoustic waves. They discussed two extreme cases: one represented isentropic unseparated flow downstream of the shock, and the second had fully separated flow (just from the shock root), the properties of which maintained constant values up to the exit. The admittance function for the excitation frequency at the channel cross-section of the time-mean position of shock wave is calculated for both cases.

Richley and Adamson,[54] Adamson and Liou,[1] Adamson *et al.*,[2] Chan and Adamson[11] and Messiter and Adamson[44,45] have developed an asymptotic solution for the unsteady two-dimensional frictionless and isentropic channel flow with shock. Adamson *et al.*[2] distinguished between two cases for the motion of the shock, resulting from a sinusoidal disturbance of the exit pressure. In the first case new shocks are formed periodically at the nozzle throat. After a short motion downstream, they run upstream

through the nozzle throat and disappear. In the other case the shock performs a periodic motion around a position downstream of the throat. These two cases are separated by the dividing case, in which the shock just reaches the throat during its upstream motion and then turns downstream. By a method similar to that of Adamson *et al.*[2] and Messiter and Adamson,[45] the unsteady flow field caused by pressure fluctuations at the nozzle exit was calculated by Liou and Sajben[34] and Liou and Coacley.[35] Here, however, the influence of an attached turbulent boundary layer is taken into account. All these investigations have in common that the pressure fluctuations considered are of low frequency and have small amplitudes only. Typically, this leads to an oscillation of the shock about an equilibrium position.

A non-linear approach to the problem of shock wave oscillations in Laval nozzle induced by the external sinusoidal excitation of exit pressure was made by Rein *et al.*[53] Their investigation was carried out numerically by the method of characteristics for cases where the shock terminating the supersonic region may be regarded as normal and no boundary layer separation occurs. For low values of frequency and amplitude, similar results, as in the above-cited papers,[1,2,44,45,54] were obtained. However, an increase to higher values led to new patterns. Additional shocks were formed which ran together with the original shock. In Fig. 23 the shock paths, resulting from pressure disturbances of different frequencies and amplitudes, are arranged side by side for comparison. The shock path starts at time $t = 0$ from the steady initial state. As soon as the first disturbance arrives at the shock, the shock location starts to change. For the lower dimensionless frequency $\bar{f} = 0.1$ (scaled by $2a_*/h_*$,

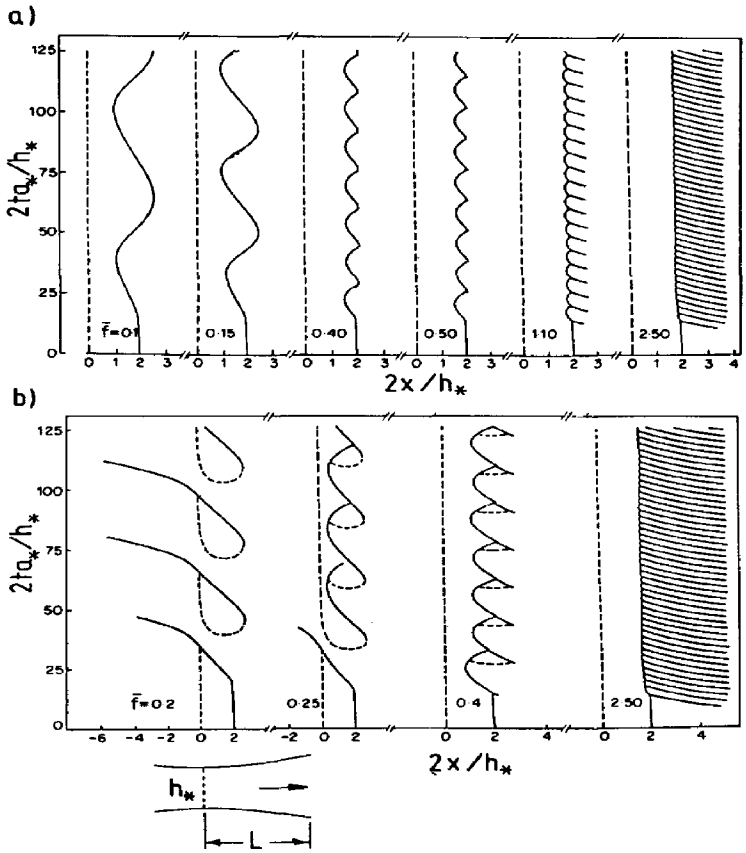


FIG. 23. Shock path resulting from exit pressure (p_e) fluctuations of different frequencies in slender Laval nozzle; $L/h_* = 3.75$, $p_0/p_e = 1.11$ (a) $\Delta p_0/p_e = 0.04$, (b) $\Delta p_0/p_e = 0.08$ (from Ref. 53).

a_*—the critical speed of sound, h_*—throat height) and amplitude $\overline{\Delta p} = 0.04$ (scaled by exit pressure p_e) (Fig. 23a), the shock path is almost sinusoidal. With increasing frequency ($\bar{f} = 0.15$) in the course of the downstream motion of the shock, a supersonic region develops behind the shock. At still higher frequencies the steepening of the pressure disturbance leads to the formation of a new shock before the disturbance has reached the original shock. Here it happens at a frequency of $\bar{f} = 0.5$. The shock path of the new shock intersects the path of the original one. A single shock of adequate shock strength results because the two intersecting shocks belong to the same family (see above). With a further increase of frequency of the excitation ($\bar{f} = 1.1$), the place of shock formation shifts downstream. This means that at high frequencies more than two shocks may exist in the nozzle simultaneously. In each period a new shock originates that needs more than one period to cover the distance from the place of formation to the original shock. This becomes clear for the frequency $\bar{f} = 2.5$.

Now, if the amplitude of the pressure disturbance is doubled to $\overline{\Delta p} = 0.08$ (Fig. 23b) with the nozzle form and exit pressure remaining unchanged, for low frequencies a quite different frequency dependence of the shock motion follows. Here, at the maximum of the disturbance, the exit pressure reaches such high values that in the steady case no supersonic flow would be possible. Thus for low frequencies, e.g. $\bar{f} = 0.2$, where the problem still may be treated quasisteadily, with increasing exit pressure the shock moves upstream through the throat. In the convergent part of the nozzle the shock strength decreases so much that the shock may be neglected in the end. The decreasing exit pressure causes an acceleration of the flow. A new supersonic region develops so that sonic conditions are not reached at the nozzle throat but somewhat downstream of it. At the downstream end of the supersonic region in the divergent part of the nozzle, not at the throat as predicted by Adamson *et al.*,[2] a new shock is formed. This one again disappears upstream through the throat due to the increasing exit pressure. After the formation of the new shock, two shocks may coexist in the nozzle again for a short while, but this time one is in the divergent and the other in the convergent part of the nozzle. At a little higher frequency, $\bar{f} = 0.25$, the flow is already accelerating before the shock reaches the throat. The shock reverses its direction of motion and coalesces with a new shock that again has formed downstream at the end of the supersonic region. Here, in the first period the shock disappears upstream because of a transient effect. With increasing frequency the supersonic region behind the original shock becomes smaller and smaller and the new shock is formed by the steepened compression wave in the subsonic region. At high frequencies, for $\bar{f} = 2.5$, the features of the shock motion for both amplitudes $\overline{\Delta p} = 0.04$ and 0.08 agree qualitatively. The higher pressure amplitude causes the new shocks to form further downstream than at small pressure amplitudes. For this reason, in the example given in Fig. 23, up to four shocks may coexist in the nozzle. This is in agreement with experiments of Weidemann[71] (see Fig. 24).



FIG. 24. Interferogram of transonic Laval nozzle flow with several weak shocks. Flow direction is from left to right (from Ref. 71).

4. SUDDEN ENLARGEMENT OF DUCT CROSS-SECTION

An abrupt increase of duct cross-section usually exists in some kinds of supersonic diffuser, ejector systems, pressure-reducing valves and exhaust valves of internal combustion engines. Large amplitude, self-excited oscillations occur frequently in such configurations in both circular and rectangular ducts.

The flow always separates at the edge of the intake duct having a lower diameter. However, it reattaches to the walls of the enlarged part at a downstream section when this part is sufficiently long.

Otherwise the flow is fully separated at least for some oscillation phases. Hence, two configurations can be distinguished: the first is a duct having a long enlarged part and the second is a duct having a short enlarged part. Below we consider both configurations.

4.1. Duct Having A Long Enlarged Cross-Section

Most of the tests have been carried out in a rectangular channel through which atmospheric air was drawn by means of a downstream vacuum. A schematic diagram, characteristic of all the ducts tested, is shown in Fig. 25.

Figure 26 shows typical interferograms obtained with a Mach–Zehnder interferometer for the different flow regimes in a rectangular duct, as the downstream receiver pressure p_e is increased. The flow structure can either be dominated by a series of reflected oblique shock waves or, when the base pressure p_w increases, by a single normal shock wave (Mach wave) close to the nozzle cross-section increase.

For very low values of exit to ambient pressure ratio p_e/p_a, the reflected oblique shock waves are repeated in the stable structure shown in photograph (a) in Fig. 26. With higher values of p_e/p_a, flow separation occurs in the presence of oblique shock waves, as is shown in photograph (b, c) where the flow separates in the region of the third and second oblique reflections respectively. In both cases the separation points oscillate over small distances. The oscillations represented by photographs (b, c) may be called the downstream (D.S.O.) and midstream (M.S.O.) oscillations respectively. A further oscillation type occurs with the higher p_e/p_a values when the main flow structure changes from an oblique to a normal shock during a cycle. Photograph (d) shows an instant during the oscillations which are of large amplitude and have been called the shock-pattern oscillation (S.P.O.). The last type of strong oscillations is the base pressure oscillation (B.P.O.) which occurs at relatively high p_e/p_a, and is characterised by strong changes of separation bubble volume.

All of the above-mentioned oscillations can be recognised in the curve of mean stagnation pressure measured at the outlet cross-section, p_{oe} (see Fig. 27). It can be noticed that the greatest decrease of the stagnation pressure appears for $p_e/p_a = 0.288$. It corresponds to a flow pattern characterised by a shock wave having the widest normal central part. In Fig. 28 this pattern is compared with another one for $p_e/p_a = 0.336$ which has a smaller central part and hence induces a smaller decrease of stagnation pressure.

In the following we give a detailed description of the shock-pattern and base pressure oscillations. These oscillations are distinguished for their strength and regularity from the other mentioned oscillation modes (M.S.O. and D.S.O.) which have a rather random character.

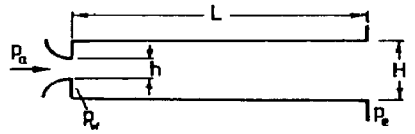


FIG. 25. Duct model.

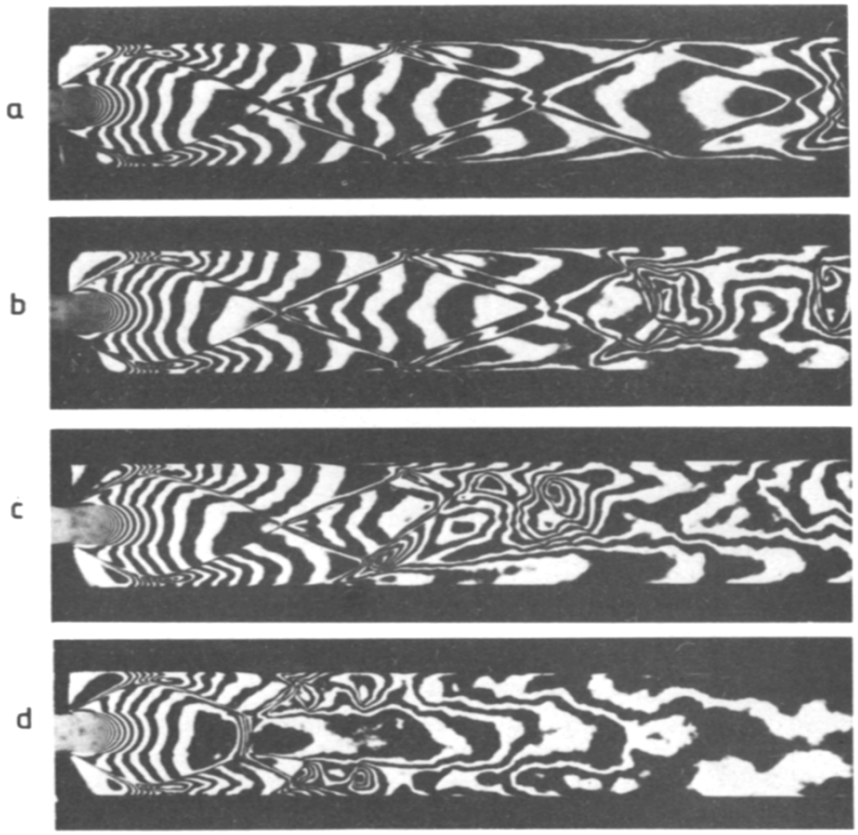


FIG. 26. Interferograms of the flow in channel having a long enlarged cross-section part. Flow direction is from left to right. $L = 160$ mm, $h/H = 0.3$, $L/H = 5.33$, (a) $p_e/p_o = 0.149$, (b) 0.189, (c) 0.211, (d) 0.264 (from Ref. 68).

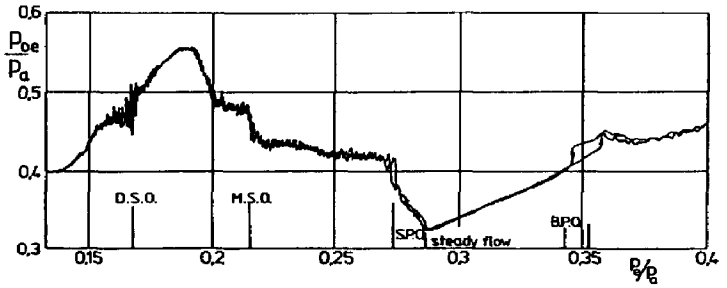


FIG. 27. Stagnation pressure measured at the outlet cross-section versus exit pressure. $L = 160$ mm, $h/H = 0.3$, $L/H = 5.33$ (from Ref. 68).

4.1.1. *Shock Pattern Oscillation*

Interferogram sequences corresponding to this type of oscillation are shown in Fig. 29. Four main phases, shown schematically in Fig. 30, have been distinguished by Szumowski and Meier.[68]

In the phase of the oscillations which is shown in Fig. 30a a symmetric Mach shock is moving upstream. The shock is accompanied by closed separation bubbles at the wall which become wider during the motion. Because the bubbles are growing unsymmetrically the end

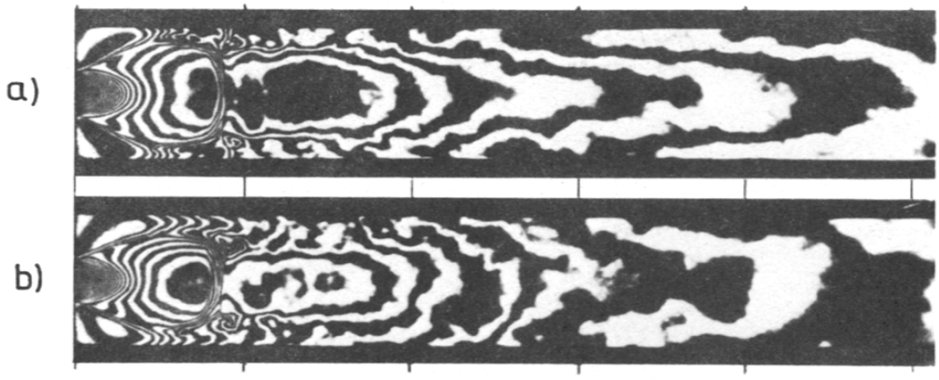


FIG. 28. Interferograms of the steady flow. Flow direction is from left to right. $L = 160$ mm, $h/H = 0.3$, $L/H = 5.33$, (a) $p_e/p_a = 0.288$, (b) 0.336 (from Ref. 68).

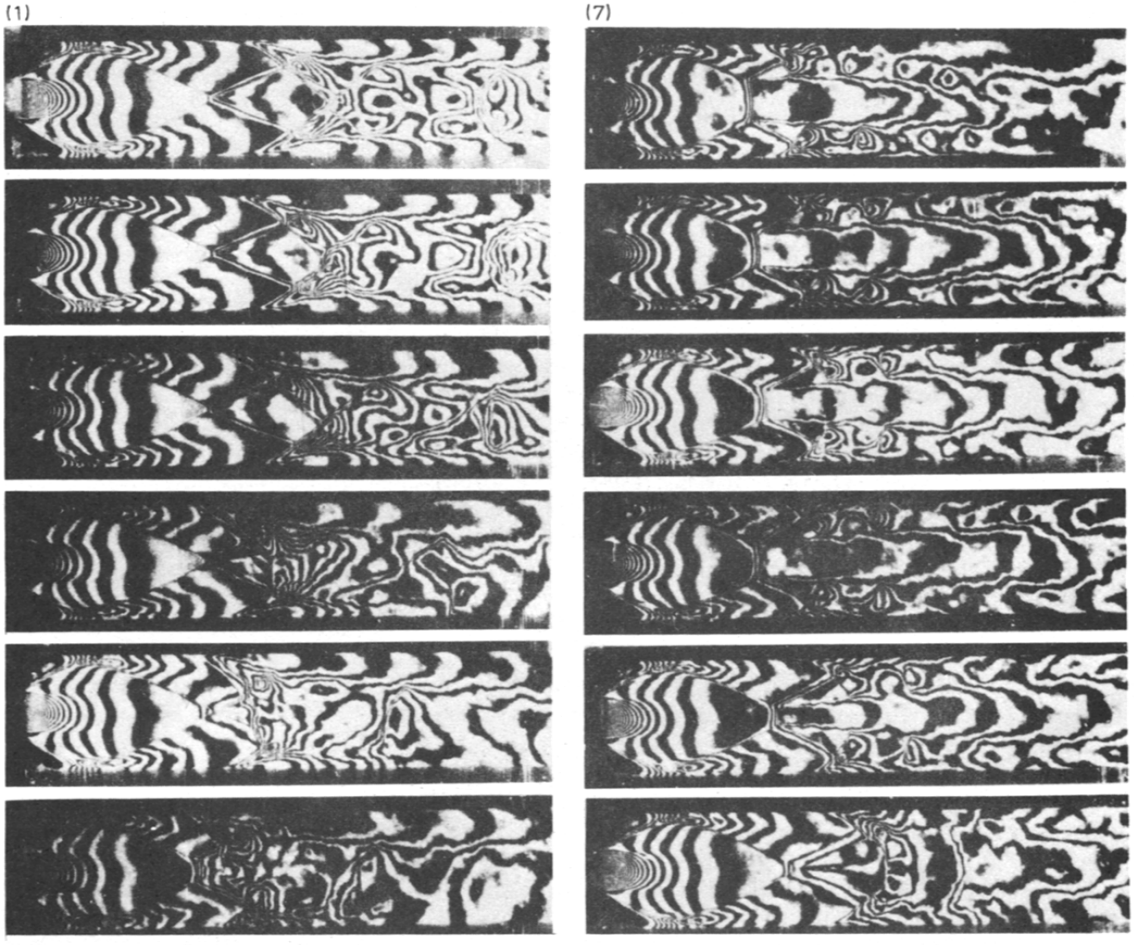


FIG. 29. Interferogram sequence of the shock pattern oscillation. Flow direction is from left to right. $L = 160$ mm $h/H = 0.3$, $L/H = 4.8$ (from Ref. 41).

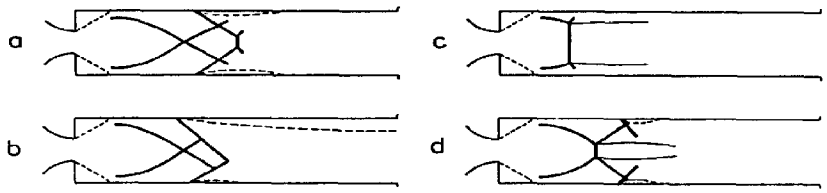


FIG. 30. Flow patterns for four main phases of oscillation cycle during the shock pattern oscillation (——) shock, (– – –) separation boundary (from Ref. 68).

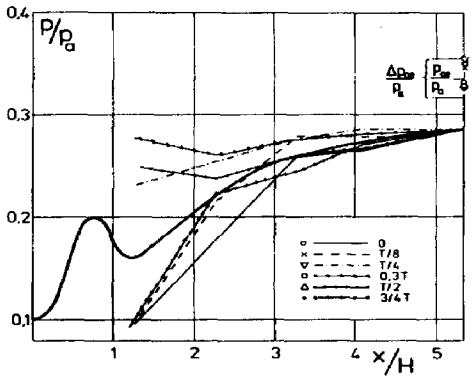


FIG. 31. Mean (thick line) and momentary pressure distributions along the duct (from Ref. 68).

shock becomes unsymmetric and the Mach configuration with triple points disappears. One of the separation bubbles, the larger one, reaches the end of the duct (Fig. 30b). To this situation corresponds the instantaneous pressure trace which is indicated with phase 0 in Fig. 31. The pressure gradient in the middle of the duct is large at this instant and the stagnation pressure at the end of the duct is nearly equal to the average stagnation pressure. Because of the connection of the separation bubble with the receiver chamber the pressure in the separation area is increased (see Fig. 31, curve $T/8$). In consequence the pressure gradient in the middle of the duct is also reduced (Fig. 31, curve $T/4$). The end shock moves upstream and a Mach shock is formed across the intersection of the inclined shock waves (Fig. 30c). The pressure curve along the duct is comparatively flat downstream of the end shock (Fig. 31, curve 0.3 T). Because of this and the narrow supersonic flow regions which are present in the vicinity of the wall, no separation occurs in the duct. This configuration is not stable because in a stationary case a larger back pressure would be needed. The relatively high pressure downstream of the end shock is explained by the fact that the extended separation bubble has reduced the pressure gradient in the duct downstream of the end shock. The subsonic core of the flow downstream of the Mach shock wave is now accelerated by the supersonic side branches of the duct flow. As a final result the pressure in this area is reduced (Fig. 31, curve $T/2$). In consequence the Mach shock is diminished and moving downstream. The pressure gradient in the area downstream of the shock increases and the separation bubbles occur at the walls (Fig. 30d). When the Mach shock has become small enough, the remaining part of the subsonic core is accelerated by the neighbouring supersonic flow regimes and disappears very quickly. Then the Mach shock at the first crossing point of the inclined shock waves disappears. However, the reflected inclined shock waves generate a new Mach shock further downstream (Fig. 30a). By this the loop is closed and the cycle of oscillation is repeated.

Szumowski and Meier[68] have observed that the shock pattern oscillation mode occurs only for such events when the mean pressure gradient near the channel exit is positive (see

Fig. 31). In this case the separation bubble being attached to the shock root easily spreads out to the duct exit and thus opens to the receiver chamber. In the case when the gradient is negative, the separation bubble remains closed. Then, only a single sudden transition from the flow pattern displayed in Fig. 30a into the flow pattern shown in Fig. 30c, and vice versa, occurs when the pressure ratio p_e/p_a slightly decreases or increases, respectively. Both flow patterns, which are characterised by pressure distributions shown in Fig. 32 are stable.

4.1.2. *Base Pressure Oscillation*

Figure 33 shows a cycle of typical base pressure oscillation in the channel of rectangular cross-section. In frame 1 the dead-air region is connected to the downstream region by a subsonic boundary layer on both sides of a central flow. The supersonic flow near the nozzle exit ends with a strong normal shock which is followed by a subsonic flow region. Supersonic layers are formed downstream of the extremities of the normal shock and surround the central subsonic region. From frame 3–6 the normal shock is moving upstream, reaching a position closest to the nozzle in frame 6. In frames 6–12 the strength of the shock wave decreases, its height is reduced and it moves downstream. Thus during the cycle the shock wave alternates in strength and the location of the shock changes, with accompanying variations in downstream flow patterns.

Additional information on shock wave behaviour during base pressure oscillation can be drawn from Schlieren photographs made by Jungowski[22] (also in Ref. 41). He used a channel of circular cross-section, supplied with compressed air. Two narrow glass windows were placed in the slits hollowed out in the channel wall on opposite sides, to enable flow visualisation. A set of frames in Fig. 34 refers to one cycle of B.P.O. The corresponding $p_w(t)$ history is shown at the side. It can be seen that at low p_w values a single strong and curved shock is present, which is seen to decay in the second frame. Then some fringes appear downstream of the main shock in frames 4–8. They are caused by the presence of the cellular structure in the outer portion of the stream. Measurements of pressure at the wall have shown that the second frame with the decaying shock corresponds to the beginning of a downstream propagation of the reattachment region.

Figure 35 displays two extremal flow patterns during B.P.O. for two values of the exit to supply pressure ratio p_e/p_0 bounding the range of the oscillation (see Fig. 27)—the upper patterns correspond to lower values of p_e/p_0. The wave pattern is alternating between on the one hand a single shock (i) and on the other either a Mach disk with cellular structure (ii) (upper pictures in Fig. 35) or a cellular structure typical of underexpanded free jets.

One can suppose that similar oscillations also occur in the diffuser flow described in Section 3. The shocklets observed in the flow by Sajben *et al.*,[56,57] depicted in the bottom picture in Fig. 19, display probably the cellular structure which appears in the outer supersonic layer surrounding the supersonic core behind the normal shock.

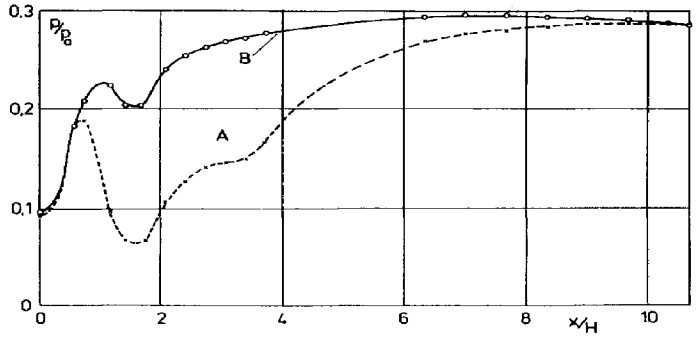


FIG. 32. Pressure distributions along the duct for two steady flow patterns: A—oblique shock waves, B—normal shock wave (from Ref. 68).

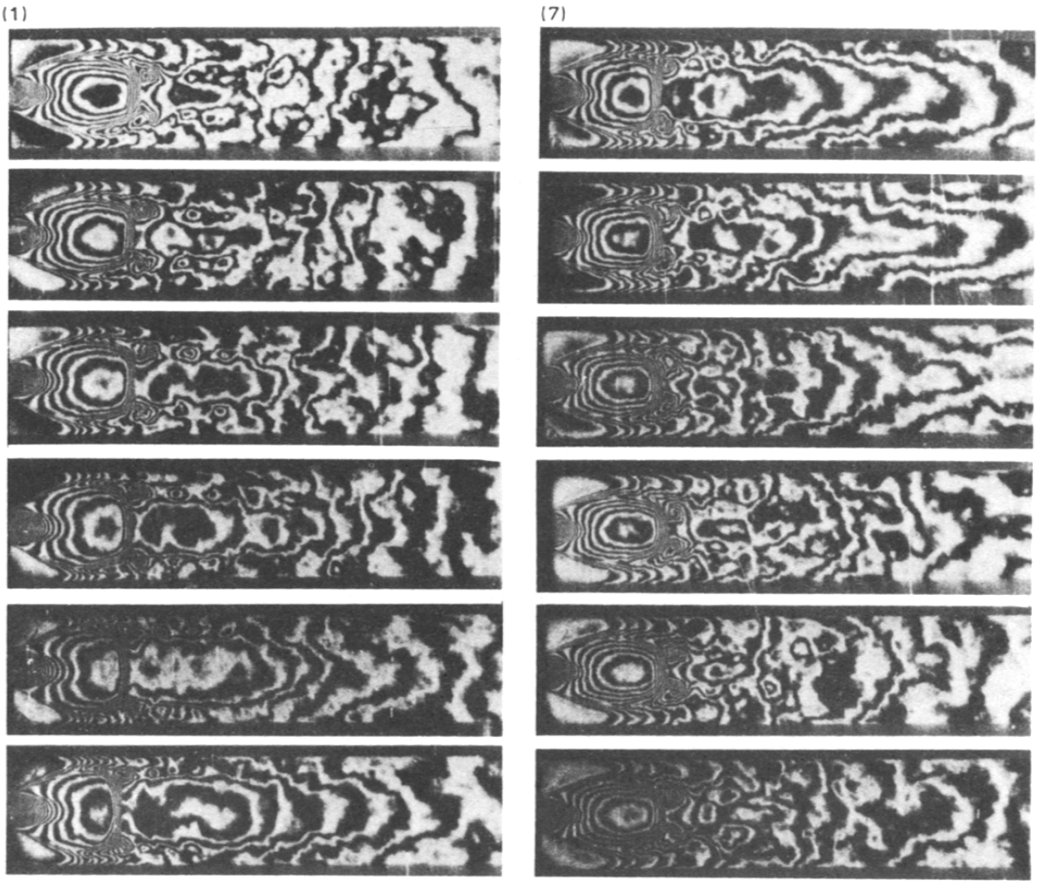


FIG. 33. Interferogram sequence at the base pressure oscillation. Flow direction is from left to right. $L = 160$ mm, $h/H = 0.3$, $L/H = 4.8$ (from Ref. 41).

Jungowski[22] concludes that the most plausible mechanism of the base pressure oscillation is as follows. The p_e/p_0 ratio range corresponding to the oscillation, the reattachment region, which secures the mass balance between air scavenged from the base region and that reversed from the reattachment region, should be located between the extreme stable positions shown in Fig. 36. The associated cellular structure, however, may be unstable in this pressure range. The stable flow was possible when a supersonic reattachment occurred with an oblique or bow shock close to the nozzle, or when a cellular structure was isolated from the wall by a thick subsonic layer which prevented the establishment of a feedback loop. The oscillation can start when a random perturbation causes the base pressure to decrease. Then the jet expands more, and air is entrained excessively from the base region to the reattachment region through the narrow slit between supersonic jet boundary and the wall of the duct. This is a self-amplifying process which comes to an end when the jet impinges on the wall and thus the reattachment region is shifted abruptly upstream. The fall of the base pressure from its equilibrium level diminishes the static pressure in the duct and leads to the deceleration of the flow. Equilibrium level refers to that level that secures a steady flow which fulfills mass, momentum and energy conservation along the duct. The emerging pressure difference at the duct end will produce in a subsonic flow a compression wave which decelerates the flow and pushes the shock into an extreme position towards the nozzle. The momentum of the expanding jet impinging on the wall may also be added to the pressure rise. In consequence the boundary layer is not able to support the increased

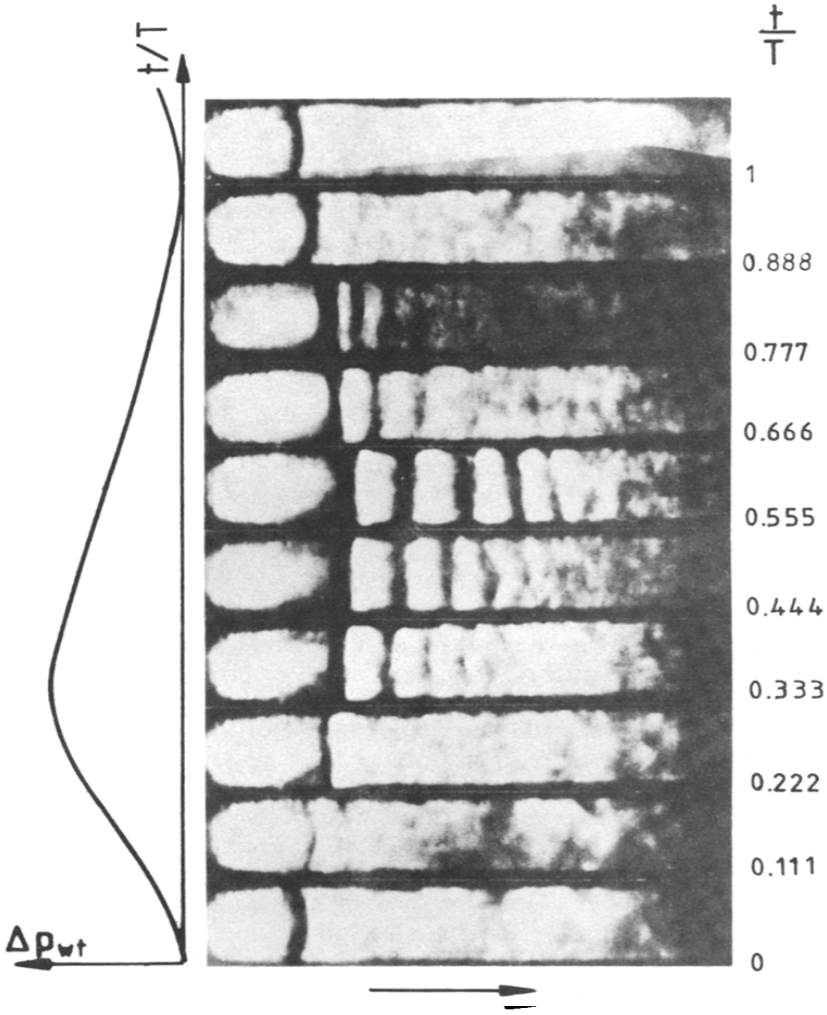


FIG. 34. One cycle of flow pattern variations during base pressure oscillation in circular duct. Interframe time $\Delta t = 0.33$ ms, $(d/D)^2 = 0.4$, $L/D = 12.7$, $D = 52$ mm (from Ref. 41).

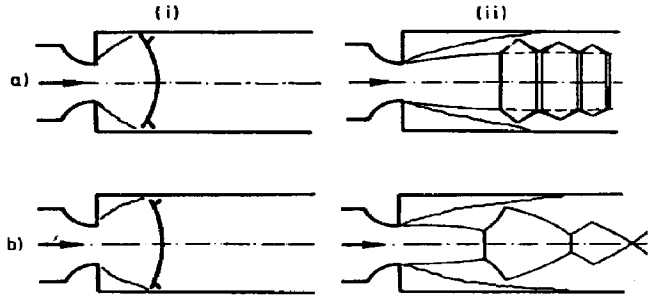


FIG. 35. Schematic diagrams of extremal (i–ii) flow patterns during base pressure oscillations. Schemes (b) correspond to lower value of supply to exit pressure ratio (from Ref. 22).

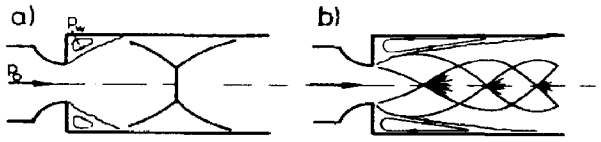


FIG. 36. Possible stable flow patterns (from Ref. 41).

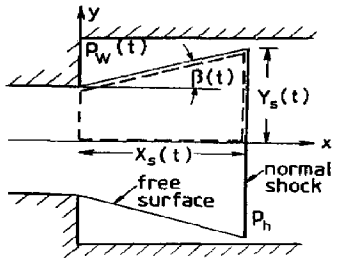


FIG. 37. Idealised flow field just behind the sudden cross-section enlargement (from Ref. 14).

pressure rise at the wall, a strong back-flow occurs and the reattachment region moves downstream. The back-flow causes a rise in the base pressure and a subsequent jet contraction at the nozzle. Consequently the static pressure increases far downstream in the duct, and then the flow is accelerated. The reappearance of the cellular structure improves the scavenging of the air from the base region, the base pressure decreases again, and the cycle is repeated. The self-supporting process of the jet divergence followed by collapse of the flow structure seems to be a very important factor in the feedback loop. The oscillation can occur in a range of frequencies, but maximum pressure amplitudes appear at a particular frequency associated with some p_e/p_o value, when probably the amplification factor is the highest. Moreover the resonance in the duct must match properly this frequency, otherwise the oscillation is weak and irregular. In a long duct the higher harmonics of resonance may be selected. The experiments and the above considerations show that base pressure oscillations depend on an interaction of a driving oscillator in the area of the base and a synchronizing resonator which is the duct.

Theoretical modelling of the oscillating flow in the considered geometrical configuration presents considerable difficulties, which arise from the complicated three-dimensional flow pattern containing shock waves, separation bubbles and both subsonic and supersonic flow regions.

An attempt at the theoretical modelling of base pressure oscillation was made by Grabitz[14] (also in Ref. 41) who considered the supersonic flow regions just downstream of the abrupt expansion terminated by a shock wave. His idealized flow field is shown in Fig. 37. Grabitz used constitutive equations of two-dimensional flow of an ideal gas in integral form for the control volume displayed in this figure, and an approximate relation between the back pressure (p_h) and the dead air pressure p_w. He took into account, in this relation, either the resonance frequency of the channel downstream of the abrupt expansion (the first version) or the secondary flow between the expansion region and the walls (the second version) assumed as one-dimensional and incompressible. Results of calculations following this theory exhibit only approximate agreement with experimental data.

4.2. DUCT HAVING A SHORT ENLARGED CROSS-SECTION

Hill and Green[19] named this configuration a 'whistler nozzle' because of its acoustical effect. They recorded a strong influence of internal flow oscillations on the free jet

characteristics which resulted in higher turbulence and faster decay of the centreline velocity. These characteristics were also investigated in detail by Hasan and Hussain[16–18] for low flow Mach number ($M < 0.2$).

Hill and Green[19] suggest that the oscillating flow in the configuration appears as an effect of cyclic separation and reattachment in the whistler lip region synchronized by the organ-pipe natural frequency. Hussain and Hasan[20] view the whistler nozzle phenomenon as a coupling of three resonance mechanisms: the shear-layer tone (an acoustic effect of interaction of the jet shear layer with downstream located lip), the organ-pipe resonance of the pipe-nozzle and the 'preferred mode' of the jet (the oscillation mode which can occur for a determined excitation frequency). They came to this view comparing the oscillating flow characteristics in three configurations (Fig. 38): whistler nozzle (a), pipe-ring (b) and pipe-hole for flow velocities $u < 60$ m/s. They found that the shear-layer tone is an important component in feedback for the pipe-ring and pipe-hole cases but has a small significance for the whistle-nozzle configuration. For this last case the shear-layer instability of the jet in the duct part of enlarged cross-section appears as the crucial factor of the oscillation mechanism (as was suggested by Hill and Green[19]).

Experiments performed for a wide range of Mach numbers, i.e. in both subsonic and supersonic flow regimes (Szumowski *et al.*,[69] Selerowicz *et al.*[62,63]) confirmed this view. Interesting findings are three oscillation modes which appeared as the supply to ambient pressure ratio was increased. The pipe-nozzle layout used in most of these experiments is shown in Fig. 39. A collar concentrically attached to the pipe formed the sudden enlargement of channel cross-section. The flow behaviour in this configuration was investigated by both base pressure and the overall sound pressure level (OSPL) measurements.

In Fig. 40 the peak to peak amplitude of base pressure oscillation is plotted against the pressure ratio p_o/p_a. In some ranges of p_o/p_a occur the strong oscillations characterised by high discrete components (Fig. 41).

The OSPL traces are shown in Fig. 42 for collar length (L_c) changes in steps of 1 mm. The peaks of all traces form characteristic ridges. The three-dimensional diagram suggests that

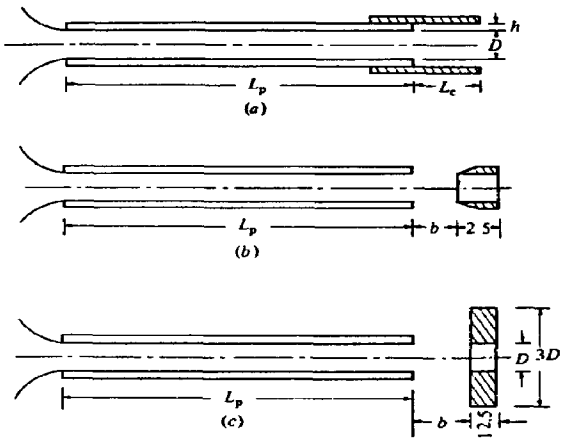


FIG. 38. Experimental configurations: (a) pipe-collar, (b) pipe-ring, (c) pipe-hole (from Ref. 20).

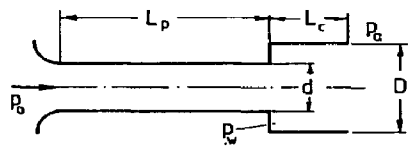


FIG. 39. Scheme of the pipe-collar nozzle.

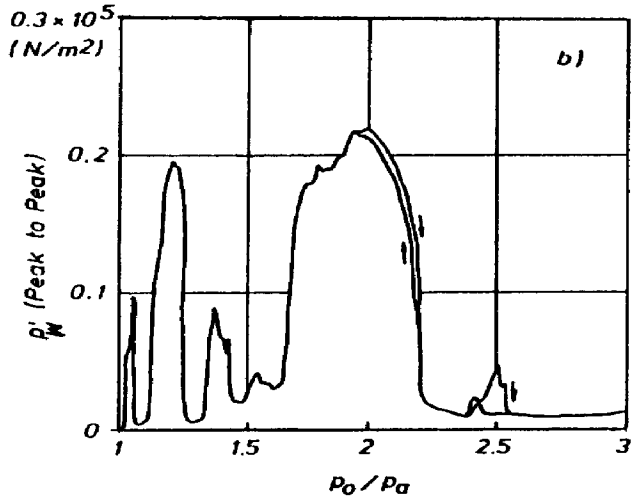


FIG. 40. Variation of base pressure amplitude with supply to ambient pressure ratio; $d = 12$ mm, $D = 18$ mm, $L_c = 10$ mm, $L_p = 60$ mm (from Ref. 63).

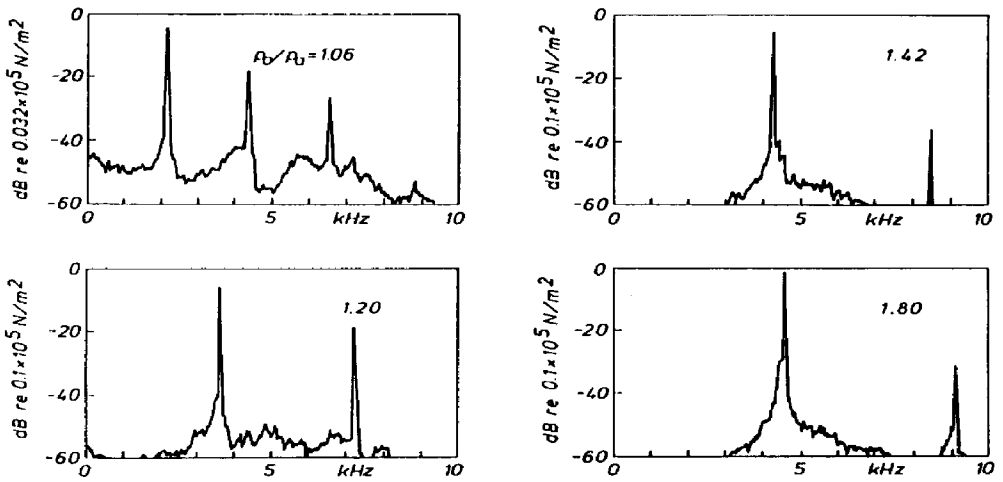


FIG. 41. Frequency spectrum of base pressure signals for four values of p_0/p_a—see Fig. 40 (from Ref. 63).

the flow oscillations corresponding to the same ridge are of the same type. The detailed measurements of the discrete frequencies as well as the flow visualisation confirm this suggestion. The results of the experiments are summarized in a L_c/d–p_0/p_a plot shown in Fig. 43. The oscillations, which belong to the regions Ia–Id are symmetrical, their frequencies correspond to the lower band shown in Fig. 44. The oscillations of region A are asymmetrical and have much higher frequencies. The oscillations of region II are, as in region I, symmetrical with frequencies in the middle band. The jet interferograms corresponding to the six regions are shown in Fig. 45. The strong toroidal vortices in the case of symmetrical oscillations and a screw vortex in the other case are visible in these photographs.

Selerowicz *et al.*[63] suggest the following most probable mechanisms of the different oscillation modes.

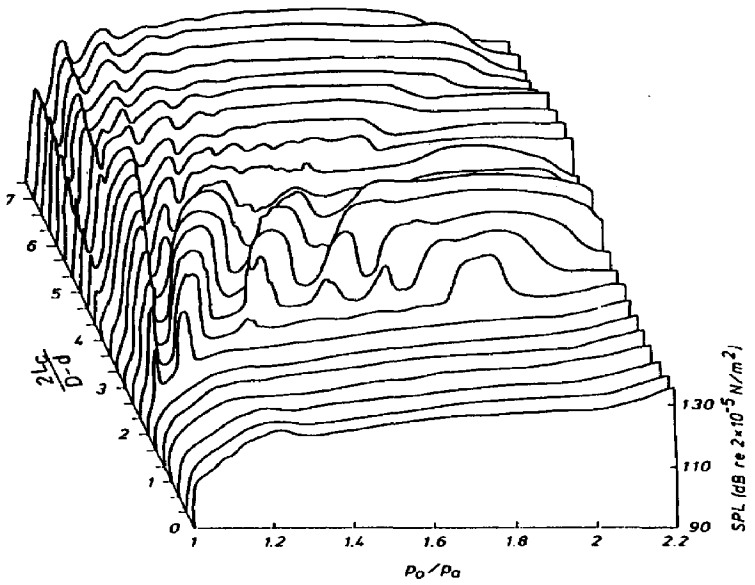


FIG. 42. Overall sound pressure level versus supply to ambient pressure ratio for various collar lengths. For pipe-collar dimensions see Fig. 40 (from Ref. 63).

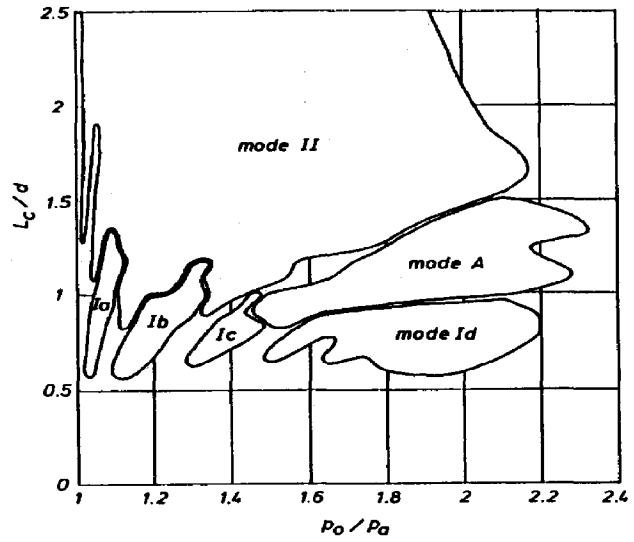


FIG. 43. Oscillation ranges in pipe-collar nozzle; $d = 12$ mm, $D = 18$ mm, $L_p = 60$ mm (from Ref. 63).

4.2.1. *Mode I (Symmetrical Oscillation)*

This mode appears for a relatively short collar. The characteristic flow patterns during the oscillation are displayed schematically in Fig. 46. We can discuss this pattern in the light of the base pressure (p_w) and flow velocity signals displayed in Fig. 47.

In the phase (i) of maximum p_w the separation bubble is open to the ambient air. The centreline velocity at the pipe outlet cross-section has a minimum whereas at the collar outlet it has a maximum (Fig. 47). About then the bubble air begins flowing out into the

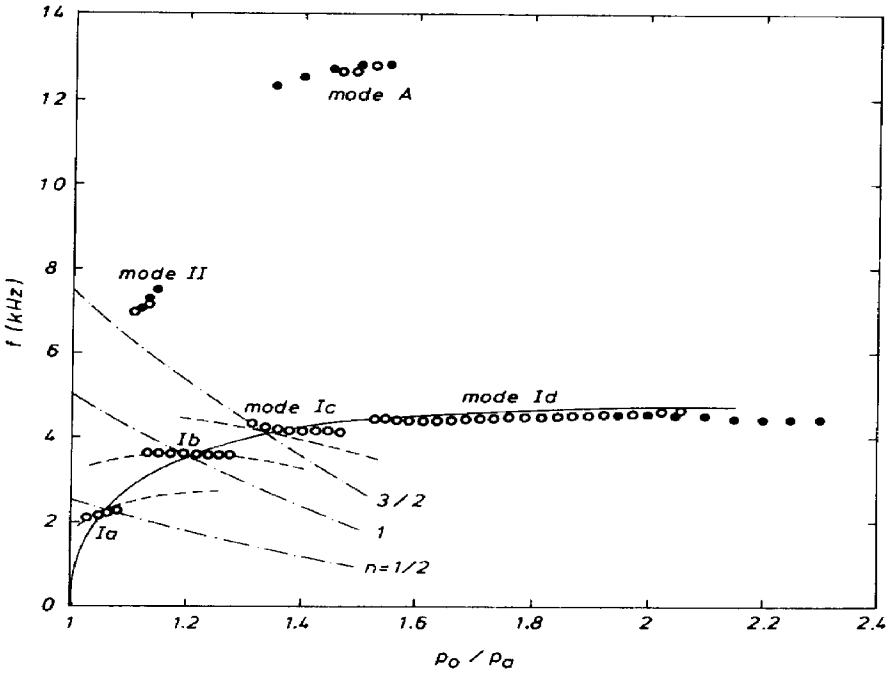


FIG. 44. Fundamental frequencies of oscillation Mode I in pipe-collar nozzle for $L_p = 60$ mm (empty points) and $L_p = 0$ (filled points) vs supply to ambient pressure ratio; $d = 12$ mm, $D = 18$ mm; (——) natural generator frequency, (—·—·) organ-pipe frequency taking into account the flow velocity (for wave length $\lambda = (0.5, 1, 1.5)\, L_p$), (— — —) average frequency of pipe-nozzle flow (from Ref. 63).

ambient air. In effect the shear layer moves outward near the collar lip (ii, iii) causing the jet cross-section to increase there; this increase is accompanied by a dropping of the centreline flow velocity at the collar outlet cross-section (v_B—Fig. 47). As the base pressure decreases the air outflow from the separation bubble ceases and the backflow develops. The backflow causes the shear layer roll-up and thus the ring vortex formation (iv). The vortex then gains strength. However it stays almost in the same position due to the momentum balance of the shear layer and back flow. This balance closes at higher values of the flow velocity near the collar outlet cross-section (v). Therefore the vortex leaves the collar (vi). Then, the separation bubble fills quickly with the ambient air and the base pressure increases. This increase is sustained due to the relatively low gas injection because of the reduction in the jet flow velocity at the outlet cross-section (v_A—Fig. 47). In this phase both the decreasing injection and gas inflow affect the base pressure. It increases to a value greater than the ambient pressure. In this way the cycle is repeated.

The crucial role in this mechanism is played by the vortex generation process inside the collar. As proposed above, the vortex generation is caused by the backflow of ambient air. The shear-layer waving caused by the base pressure oscillation may also be of some importance in this process. It seems however, that for great flow velocities this effect is a second order factor in the vortex formation because the vortex appears at a relatively short distance downstream of the pipe outlet cross-section.

This oscillation mechanism has been confirmed by flow visualisation in a rectangular channel (Fig. 48). Because of the small density gradient in their experiments the flow pattern is not so distinct as in the photographs shown in Section 2. However, the vortex is visible in its originating phase (Fig. 46) (iv). The backflow at the collar wall can also be noticed.

In order to check the organ-pipe frequency influence on the oscillation mechanism, Selerowicz *et al.*[63] have done some experiments with collar-nozzle configurations; in this

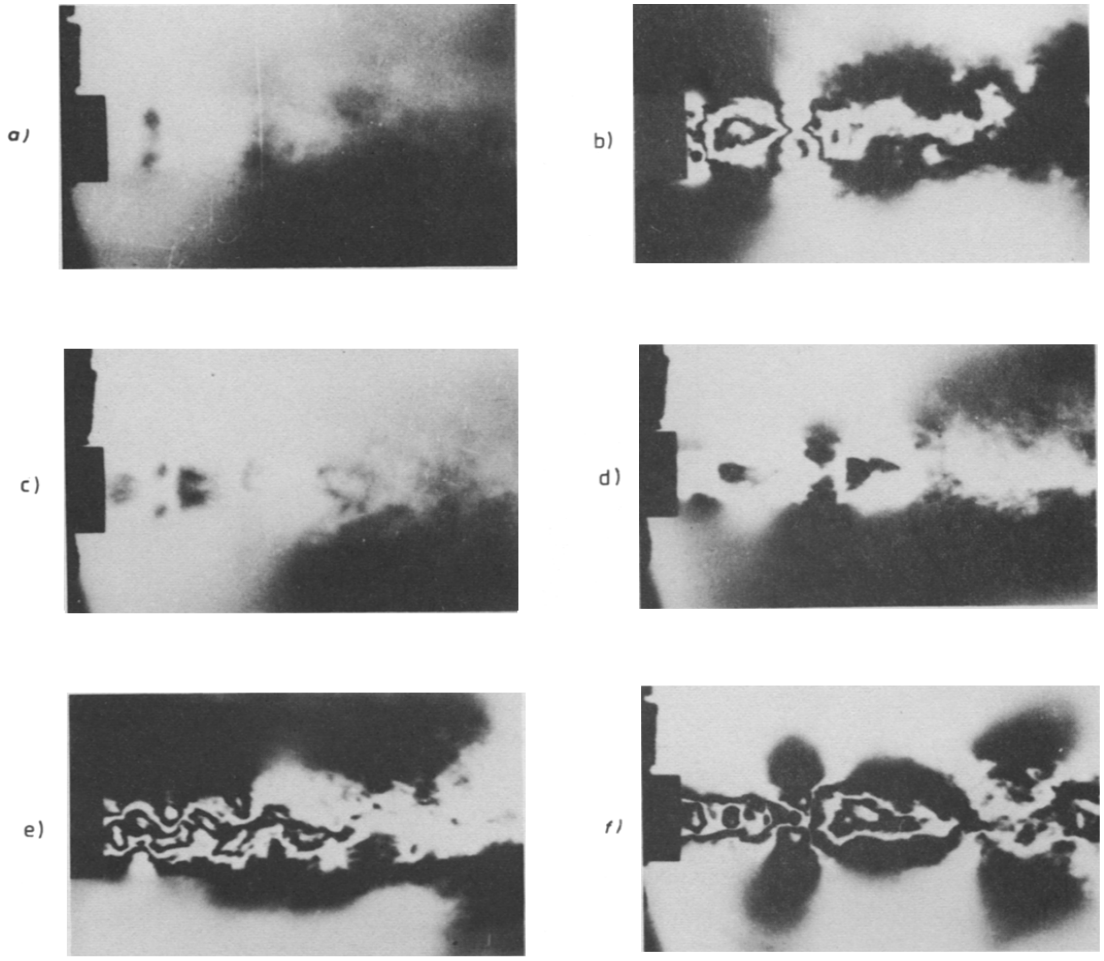


FIG. 45. Interferograms of free jet from pipe-collar nozzle; (a) Mode Ia, $p_o/p_a = 1.05$; (b) Mode II, 1.12; (c) Mode Ib, 1.20; (d) Mode Ic, 1.40; (e) Mode A, 1.50; (f) Mode Id, 2.0 (from Ref. 63). For pipe-collar dimensions see Fig. 40.

case the pipe between the nozzle and the collar was removed. The oscillation frequencies corresponding to this case are marked in Fig. 44 by the filled points. Note that the oscillations which belong to the Mode I disappear for low flow velocities, but they occur in the transonic flow regime only. This case however has not been investigated further.

One can conclude that in the subsonic flow range for relatively low Mach numbers the organ-pipe resonance seems to be an important factor. In this case the oscillation frequencies of pipe-collar-nozzle flow (dashed lines) appear as mean values of the natural generator frequencies (the continuous line in Fig. 44 is drawn through the centre points (maximum amplitudes) of the segments Ia–Id) and the organ-pipe frequencies are shown for wave lengths $\lambda = (2, 1.5, 1, 0.5)\, L_c$ (chain-dotted lines).

4.2.2. *Mode II (Symmetrical Oscillation)*

This mode occurs when the collar is a little longer (see Fig. 43). The interesting characteristic feature of these oscillations is a non-monotonic variation of the wall pressure

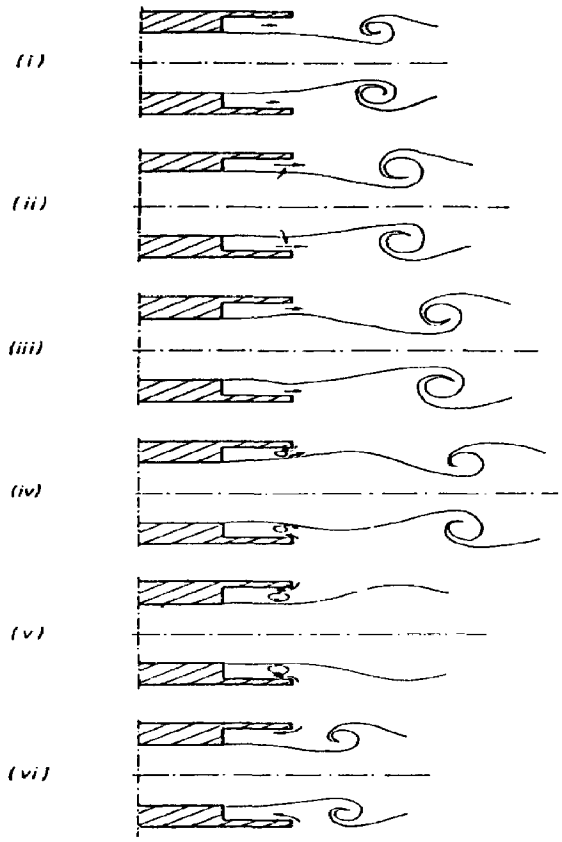


FIG. 46. Schemes of flow patterns during the oscillation Mode I; phase difference between succeeding stages $\phi/2\pi = 1/6$ (from Ref. 63).

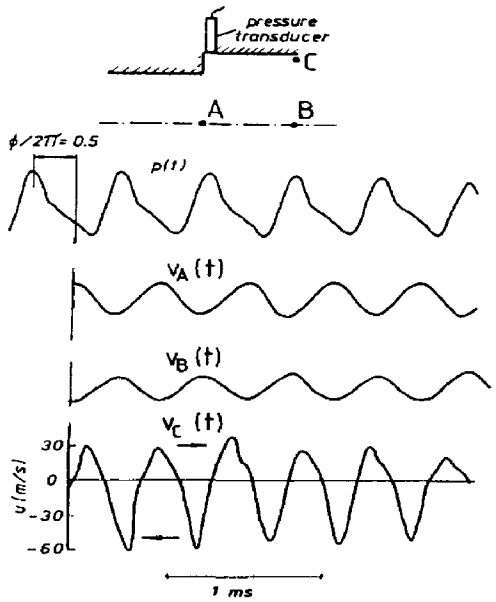


FIG. 47. Base pressure and flow velocity traces for oscillation Mode I (from Ref. 63). For pipe-collar dimensions see Fig. 40.

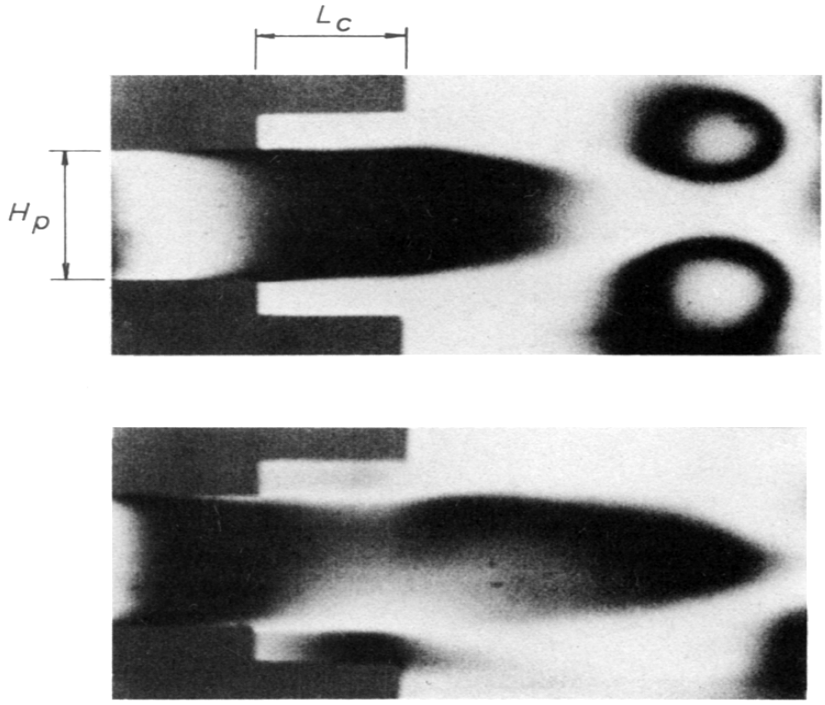


FIG. 48. Flow interferograms during oscillation Mode I for two extremal phases; flow direction is from left to right (from Ref. 63).

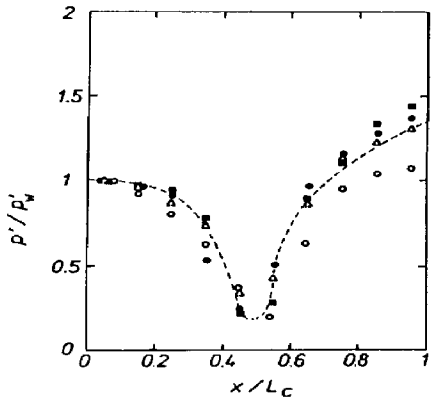


FIG. 49. Wall pressure amplitude distribution along the collar normalized by base pressure amplitude for oscillation Mode II; $d = 12$ mm, $D = 18$ mm, $L_c = 20$ mm, $L_p = 60$ mm (from Ref. 63).

signal amplitude with distance along the collar—Fig. 49 (for mode I the amplitude decreases slightly). This variation is independent of the flow velocity under consideration and shows a deep minimum at approximately half of the collar length. The phase variation of pressure signals shown in Fig. 50 displays high phase velocity in the collar part close to the cross-section enlargement and slow phase velocity in the remaining part. The variations displayed in Figs 49 and 50 lead to the instantaneous pressure distributions along the collar, shown in Fig. 51. These distributions as well as the interferometric photographs of the flow

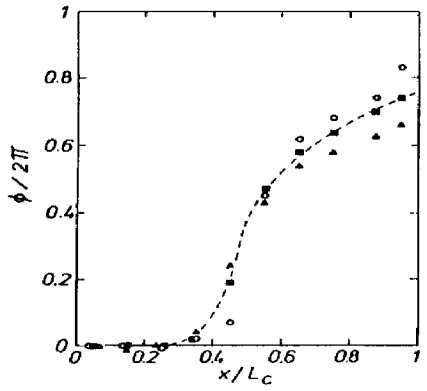


FIG. 50. Phase variation of pressure signals for oscillation Mode II (from Ref. 63). For pipe-collar dimensions see Fig. 49.

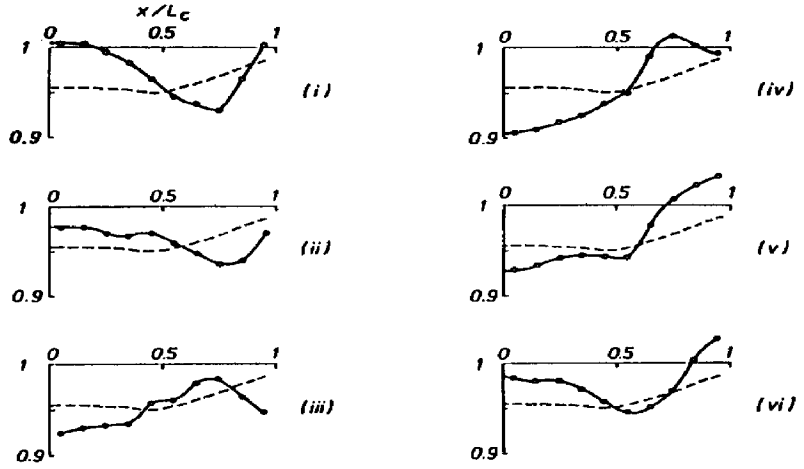


FIG. 51. Instantaneous wall pressure distributions along the collar during oscillation Mode II; phase difference between succeeding stages $\phi/2\pi = 1/6$ (from Ref. 63). For pipe-nozzle dimensions see Fig. 49.

in the rectangular duct (Fig. 52) substantiate the following oscillation flow behaviour for Mode II.

We start our description with the reference phase ($\phi = 0$) corresponding to the pressure distribution, flow photograph and schematic flow pattern shown in the first (i) diagram or photograph in Figs 51–53, respectively. In this phase the base pressure is a maximum. The wall pressure decreases along the collar, achieves the minimum value at x/L_c of about 0.8 and increases again. The interferometric photography shows a strong vortex inside the collar, in this phase. The vortex moves downstream (ii—Figs 52 and 53); this motion is accompanied by the shift of the low pressure region (ii—Fig. 51). In phase $\phi/2\pi = 0.33$ (iii) the vortex leaves the collar and the wall pressure close to the collar lip has a minimum value. The base pressure decreases and the jet (behind the vortex) approaches the collar wall. In phase $\phi/2\pi = 0.5$ (iv) the high pressure region rebuilds at x/L_c of about 0.7. The interferometric photography corresponding to this phase indicates flow reattachment at this point and a new vortex formation. At that stage, both the separation bubble and base pressure are minima. In the following phase (v) the flow near the outlet cross-section of the pipe accelerates. In effect the stagnation point and accompanying vortex shift downstream

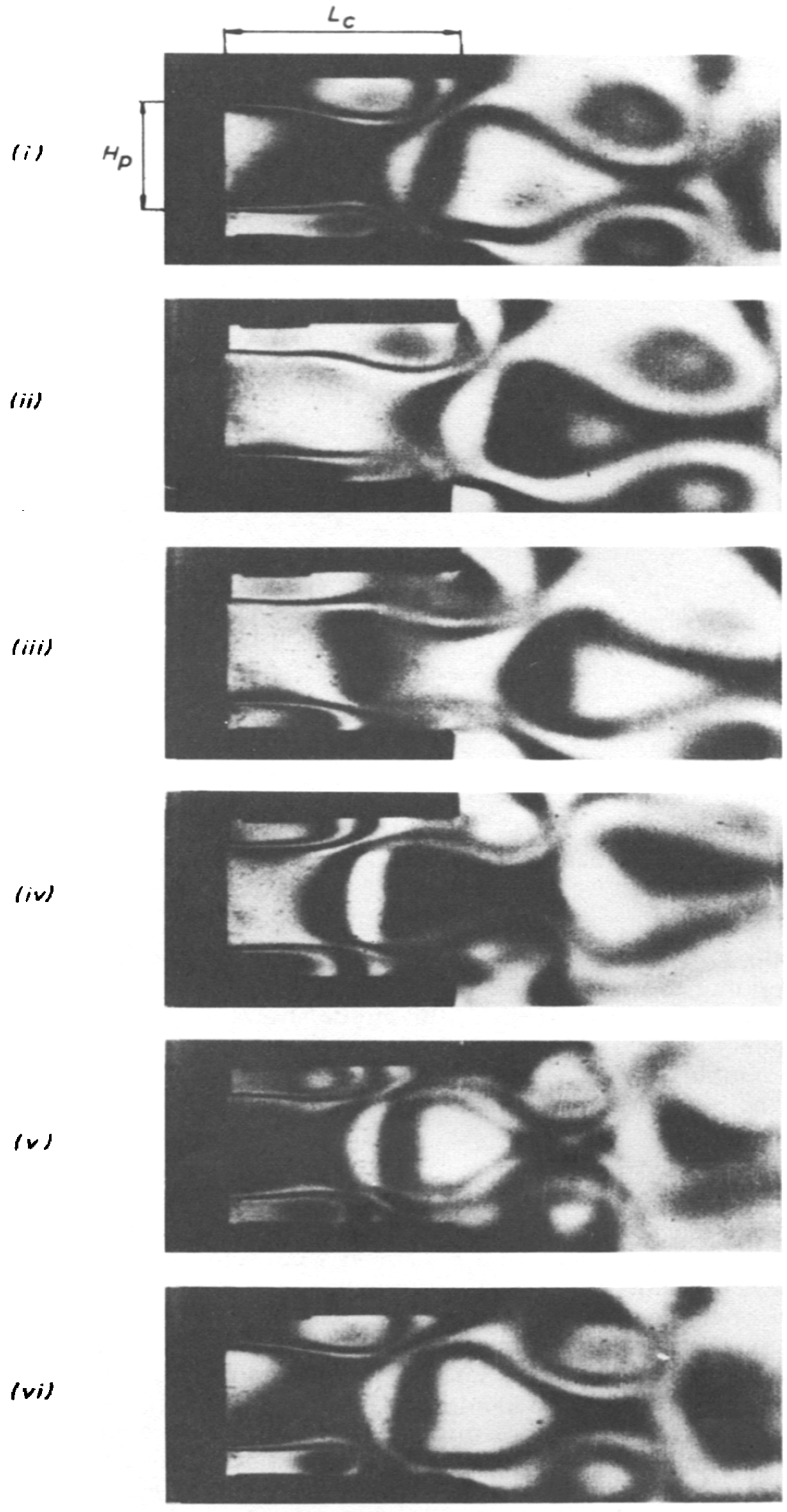


FIG. 52. Flow interferograms during oscillation Mode II; phase difference between succeeding stages $\phi/2\pi = 1/6$ (from Ref. 63). Flow direction is from left to right. For pipe-nozzle dimensions see Fig. 49.

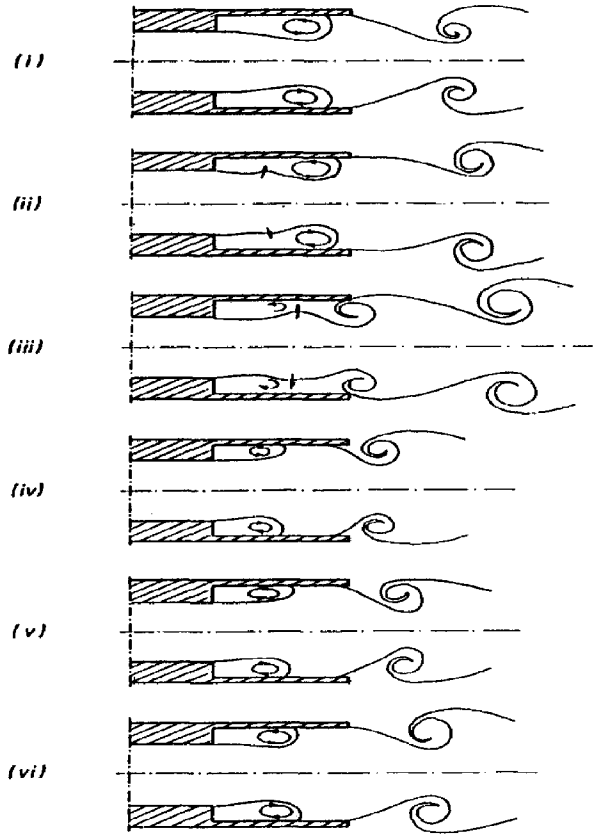


FIG. 53. Schemes of flow patterns during oscillation Mode Ii; phase difference between succeeding stages $\phi/2\pi = 1/6$ (from Ref. 63). For pipe-nozzle dimensions see Fig. 49.

up to the collar lip. Eventually, the separation bubble opens to the ambient air and the flow pattern corresponding to the reference phase rebuilds.

In contrast to the oscillation Mode I, this mode shows very weak dependency on the organ pipe resonance (Fig. 54). It seems that the oscillation Mode II is controlled most of all by the separation bubble, which periodically changes its volume. In this sense the oscillation Mode II is like the base pressure oscillation described in Section 4.1.2.

4.2.3. *Mode A (Asymmetric Oscillation)*

The asymmetric mode seems to be more difficult to explain. The base pressure measurements show that during this oscillation p_w is nearly constant. The flow velocity measurements along the centre-line, taken inside the collar, also do not exhibit any oscillation. Oscillations were observed beyond the collar exit in the free jet; however the participation of the discrete components in the velocity oscillation spectrum was very small. This proves that the spiral vortex, displaced in the jet mixing layer does not influence the jet core. The oscillation can therefore be made evident only by means of flow visualisation and acoustic measurements. The structure of the spiral vortex, observed by means of visualisation, suggests that the vortex originates at the pipe-exit edge. The pictures of the successive stages show furthermore that the starting point moves around the edge with the oscillation frequency.

The passage from the symmetric mode to the asymmetric one seems to be connected with the radial resonance frequency of the pipe. For an incompressible flow the first radial

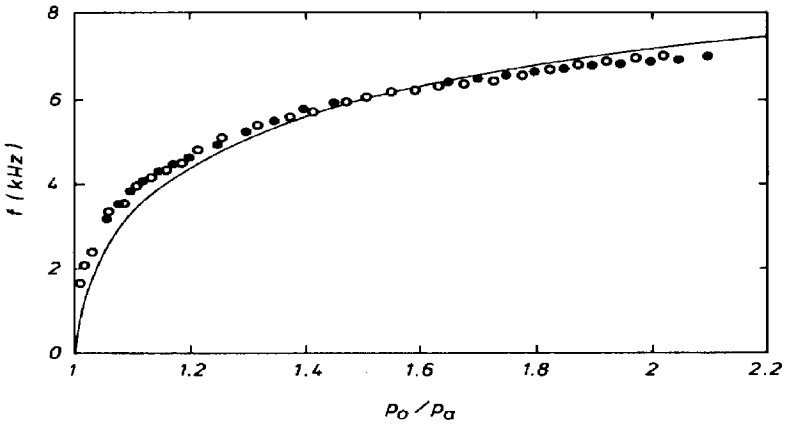


FIG. 54. Fundamental frequencies of oscillation Mode II in pipe-collar nozzle for $L_p = 60$ mm (empty points) and $L_p = 0$ (filled points), $d = 12$ mm, $D = 18$ mm, $L_c = 20$ mm (from Ref. 63).

resonance frequency can be calculated as (Skudrzyk[64]) $f = 0.293\ a/R$ where a denotes the sound speed and R is the pipe radius. For $a = 340$ m/s and $R = 6$ mm, $f = 16.6$ kHz. The experimental data of Selerowicz *et al.*[63] show that the jump from the symmetric to the asymmetric mode occurs, when the oscillation frequency exceeds a value of about 8.5 kHz, i.e. about half the value calculated above. This difference can be due to compressibility effects, so that a term of the form $(1 - M)^2$ should be taken into account. A detailed experimental study of this mode of excitation seems to be desirable.

4.2.4. *Supersonic Flow Case*

However, for high pressure ratios p_0/p_a the flow inside the pipe-collar nozzle remained stable; we will briefly discuss this case because of its practical interest. Anderson and Williams[3] observed that with a continuously varying supply pressure (p_0) the noise level emitted by the jet from such a nozzle changed suddenly by about 15 db at some values of p_0. The jet noise dropped suddenly at a certain $p_0 = p_{0d}$ while the supply pressure was increased and again jumped rapidly back to a higher level, say $p_0 = p_{0i}$, when p_0 was reduced. It has been established by Anderson and Williams[3] that this phenomenon was connected with flow separation in the enlarged part of the nozzle (the collar). The noise level decreased when the jet initially separated reattached to the collar and increased when it separated again. The experiments showed that the reattachment occurred at a higher supply pressure than the separation ($p_{0d} > p_{0i}$). Detailed investigations done by Meier *et al.*[42] indicated that shock wave changes were the reason for this phenomenon. These changes can be deduced from schlieren photographs of the jet from the pipe-collar nozzle, shown in Fig. 55. The variation of the mean value of the ratio of the base pressure (p_w) to the ambient pressure (p_0) with the ratio of the supply to ambient pressure is also displayed in this figure.

The separated slightly underexpanded jet characterised by a cellular structure can be observed at the point a in Fig. 55 (photograph a). When p_0 was increased to point b the jet expanded continuously and a Mach shock wave appeared at its first cell (photograph b). Under these conditions, with a very small increase of p_0 the jet attached to the ring wall. It was accompanied by a sudden decrease of p_w (point c) and a rapid change of the flow pattern (photograph c). This new pattern was characterised by the appearance of a strong bow shock wave in the central part of the jet. The wide subsonic core behind this shock was surrounded by a thin supersonic layer. This pattern changed slightly with further increase of p_0 (photograph d). The subsonic core reached its maximum diameter when p_0 was decreased again to a value corresponding to point e, just before the jet separation (photograph e). In

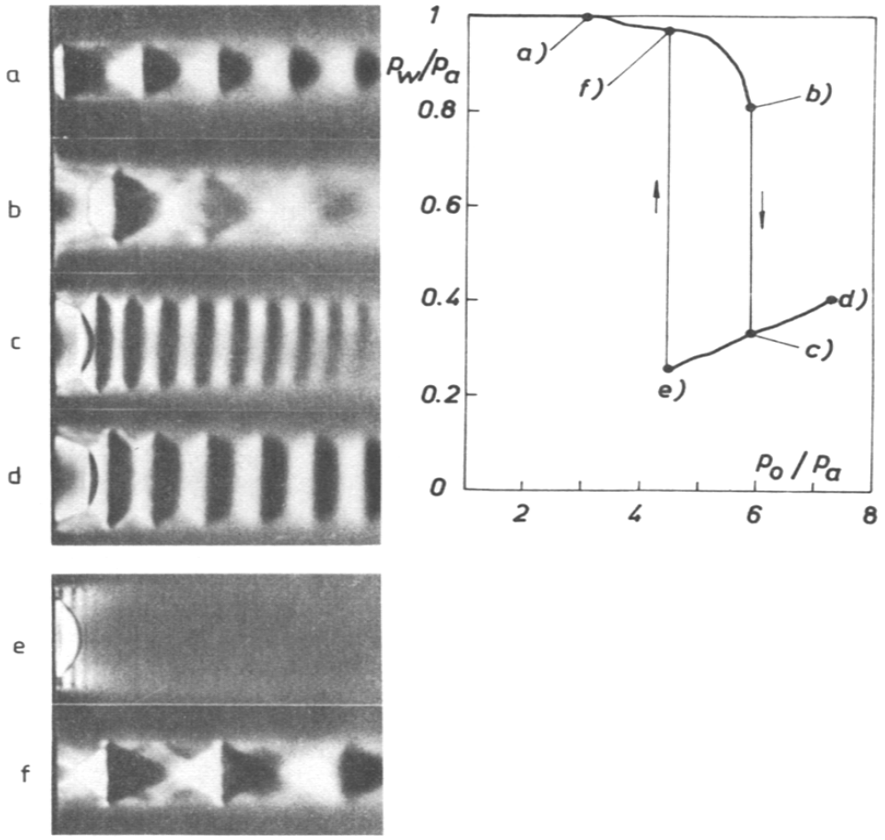


FIG. 55. Base pressure vs supply pressure variation and corresponding jet photographs; $(d/D)^2 = 0.3$, $L_c/D = 0.54$ (from Ref. 42).

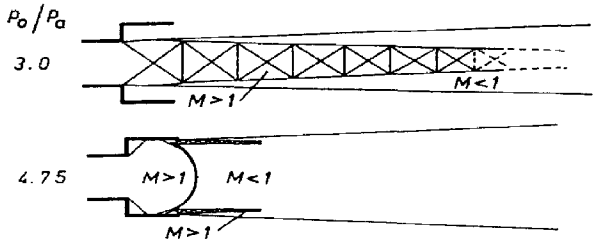


FIG. 56. Schemes of two flow patterns corresponding to photographs (a) and (e) in Fig. 55 (from Ref. 42).

this case, the supersonic layer became very thin and decayed over a short distance close to the exhaust cross-section. Then, a very small decrease of p_o caused the jet separation. This resulted in a free jet flow pattern having a typical cellular structure (photograph f). The extremal flow patterns corresponding to photographs (a) and (e) are schematically displayed in Fig. 56. The overall sound pressure level (30 cm from the jet in the plane of the outlet cross-section) versus the pressure ratio p_o/p_a is shown in Fig. 57 for the case of a short and a long collar. In the second case the OSPL changed in two steps when the supply pressure increased or decreased. The intermediate step corresponds to the base pressure oscillations described in Section 4.1.2. For the subsonic flow regime, the noise level of the jet from the

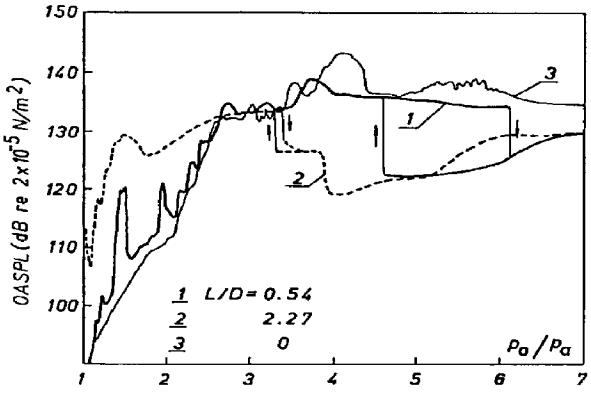


FIG. 57. Overall sound pressure level variations for various collar length L_p = 60 mm, d = 12 mm, D = 22 mm (from Ref. 42).

nozzle under consideration was greater than the one from the duct without the collar. It is caused by the flow oscillations described in Sections 4.2.1–4.2.3.

5. JET–CAVITY INTERACTION

The flow in the jet–cavity interaction exhibits two characteristic peculiarities, which have some practical importance: high intensity sonic and ultrasonic wave generation and intense heating of the gas stagnant in the tube.

The oscillation in a cavity was first described by Hartmann.[15] He observed that high-intensity sound waves were generated when an air-jet oscillator such as a resonance tube was placed in the region where the pressure in the jet is rising, i.e. the so-called 'zone of instability'.

Sprenger[67] studied at length the effects which nozzle-tube spacing, nozzle stagnation pressure, and length of the tube have on the endwall tube temperature. For a given resonance tube and jet flow, many peaks in the endwall tube temperature about as high as 750 K were observed.

It was also found that, for a given resonance tube, large variations in endwall gas temperature resulted when slight changes in nozzle pressure ratio or spacing between the nozzle exit and the tube inlet were made.

Some of the main forms of Hartmann–Sprenger generators are shown in Fig. 58. They were driven by supersonic and choked jets issuing from a convergent (a) or a Laval nozzle (b). The generators usually consist of a single resonator (a,b,c—Iwamoto *et al.*,[21] Kawahashi *et al.*,[29,30] Neemeh *et al.*;[47] g—Jungowski and Meier,[25] Jungowski and Grabitz;[26] h—Wu *et al.*[72,73]). But sometimes a second one was mounted coaxially (d—Kawahashi and Suzuki,[28] Brocher and Pinna[10]) or perpendicularly (e—Kawahashi and Suzuki;[31] f—Jungowski and Sobieraj[24]). An open tube section terminated by an anechoic end (f), a coaxial disc on the primary resonator or a horn can be applied as a secondary resonator. This resonator can, if tuned properly, strengthen or weaken the oscillation in a certain range of frequency. A tapered tube (c) or a cylindrical resonator (h) increased maximum gas temperature in the device considerably. Very low-frequency oscillations (of the order of 0.5 c/s, or less) may be obtained by using a large Helmholtz resonator (g) for the cavity. The configurations (b–h) were used in order to intensify or modify the acoustical and thermal effects of the Hartmann–Sprenger generator. The investigations of these configurations were focused on the engineering applications rather than on the study of flow phenomena. Since this review concerns the latter we consider here only the base configuration shown in Fig. 58a.

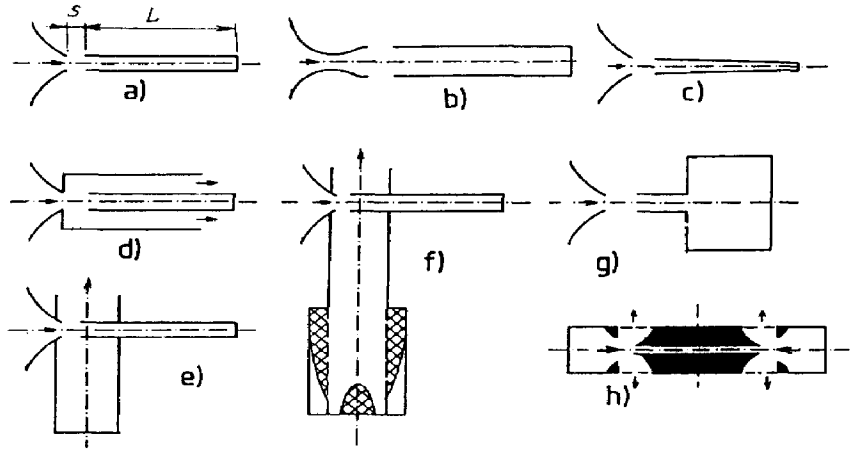


FIG. 58. Forms of Hartmann–Sprenger generator.

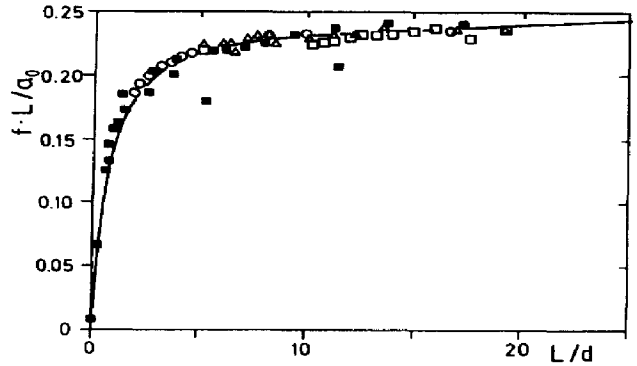


FIG. 59. Dimensionless frequency upon the nozzle-tube spacing for four nozzle (tube) diameter d: □—16 mm, △—12 mm, ■—6 mm, ○—3 mm (from Ref. 66).

Smith and Powell[65] analysed and clarified some shortcomings of Hartmann's theory, largely by identifying the bi-stable condition of the stagnation interface. The position of the interface and the stagnation pressure are coupled with the position of the detached shock wave along the jet.

Smith and Powell[65] as well as Sarohia and Back[61] distinguished two modes of operation of Hartmann–Sprenger (HS) generator in the case of an underexpanded jet from a convergent nozzle namely the jet regurgitant mode and the jet screech mode.

5.1. The Jet Regurgitant Mode

The jet regurgitant mode consists of periodic swallowing and discharging of the jet flow by the tube at a frequency, which depends on the supply to ambient pressure ratio and the system dimensions. For a moderately underexpanded jet and relatively long tubes, this mode shows approximately the fundamental tube resonance frequency, which corresponds in this case to a nearly constant Strouhal number $Sh = fL/a_0$ (Fig. 59), where L is the tube length, a_0 is the sound velocity.

A slightly modified form of the wave diagram for the flow inside a long tube proposed by Thompson[70] for the regurgitant mode is displayed in Fig. 60. The first part of the

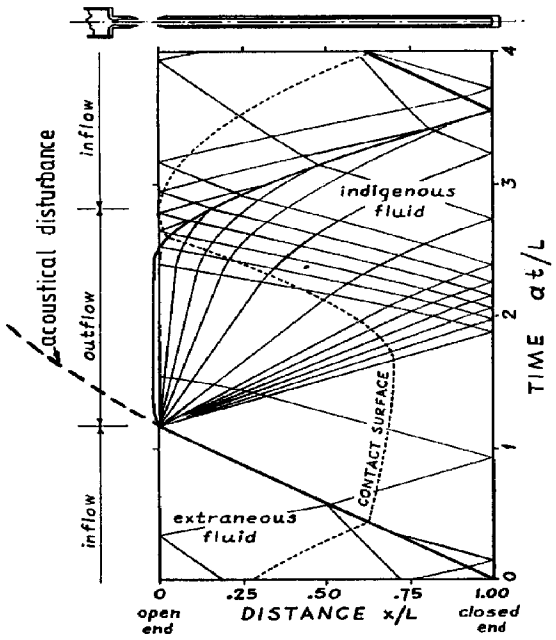


FIG. 60. Wave diagram for regurgitant oscillation mode (from Ref. 70).

regurgitant mode of tube operation is the inflow phase of the tube. Most of the jet flow enters the tube. The jet shock wave structure in this phase of the cycle depends only on the nozzle pressure ratio. That is, the jet flow between the nozzle exit and tube inlet behaves as if the tube is not present in the jet flow. This behaviour is visible very distinctly in photographs 1 and 8 in Fig. 61 showing the external flow patterns, visualised by the shadograph method. The portion of the jet flow that enters the tube causes the formation of compression waves. These compression waves then travel toward the endwall. Coalescence of the compression wave train to a shock wave depends on the tube length (Fig. 60). The waves are reflected from the endwall as a shock wave or a shock wave followed by a train of compression waves. The strengthening of the shock wave by coalescence of compression waves continues until this reflected wave front reaches the inlet of the tube. From there the compression wave front results in the formation of a strong expansion wave front which travels into the tube. The flow then suddenly switches to a transition phase followed by the outflow phase of its cycle in the regurgitant mode. The presence of the expansion waves and reduced pressure behind the wave front causes the gas in the tube to flow outwards. The transition from the inflow to the outflow phase results in a collision of the nozzle jet and the tube-jet flow, thereby forming an interface (photographs 4 and 5 in Fig. 61). As the tube-jet flow gains in strength, the plane of the interface moves toward the nozzle. At the end of the transition period between the inflow and the outflow phases, the two opposing flows remain in this stage almost for the same period of time as the inflow phase of the jet regurgitant mode. The outflow phase of the tube is completed as the expansion fan approaches the tube inlet after reflection from the endwall. This results in a weakening of the tube-jet flow. In effect, the interface quickly moves towards the inlet of the tube. Then the nozzle flow again reverts back to the inflow phase of the regurgitant mode, and thus one cycle of operation is completed.

The transition from the inflow phase to the outflow phase is very short and is accompanied by a strong acoustical disturbance (a shock wave) which spreads out in the ambient air (Fig. 62).

The theoretical treatment of the resonance tube flow during the regurgitant oscillation mode is, in substance, correct, although the one-dimensional approach is used

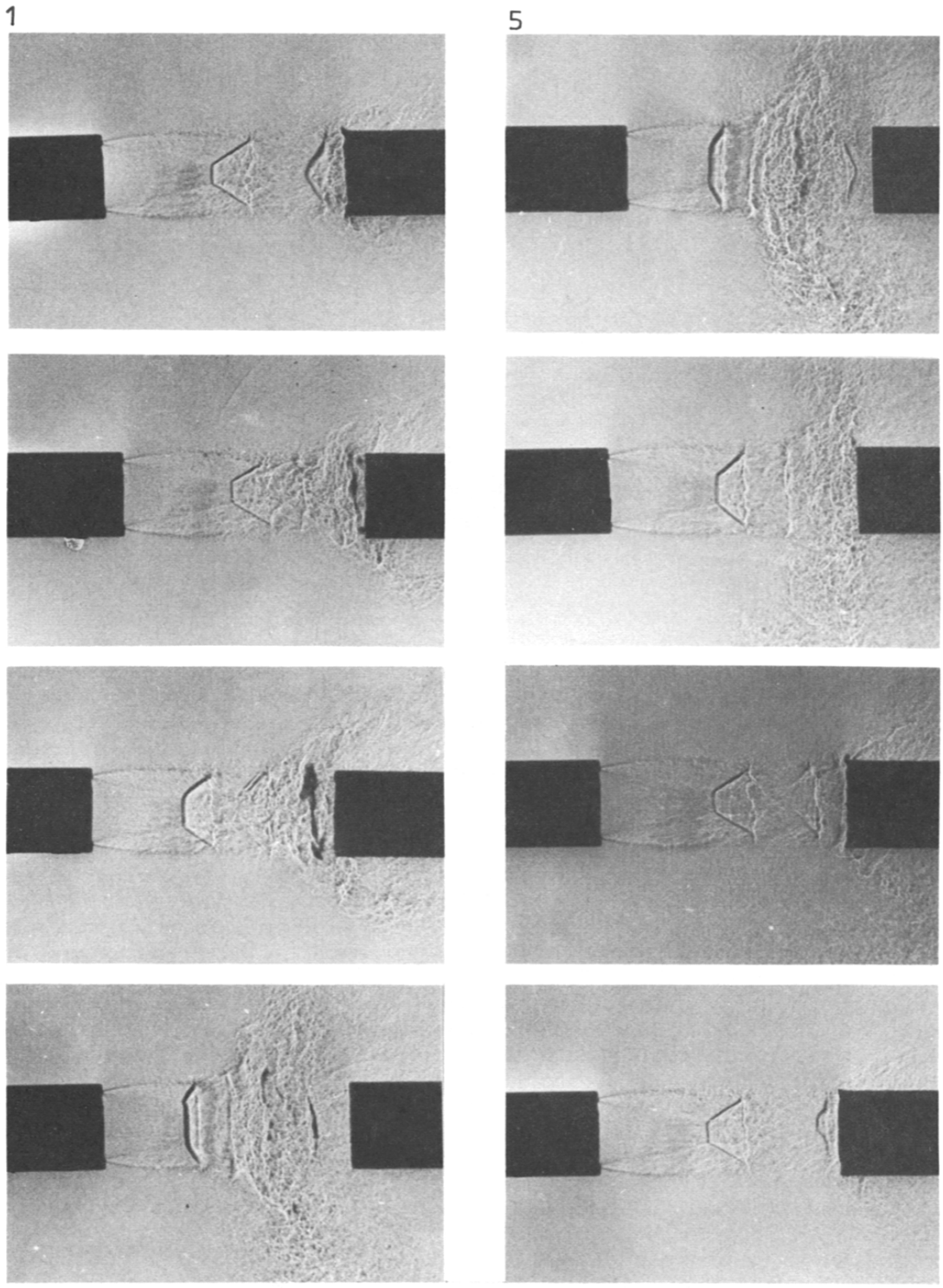


FIG. 61. External flow patterns during regurgitant oscillation mode; flow direction is from left to right. $d = 8$ mm, $L = 50$ mm, $s = 23.5$ mm, $p_o/p_a = 5.0$, fundamental frequency $f = 1.28$ kHz (from Ref. 66).

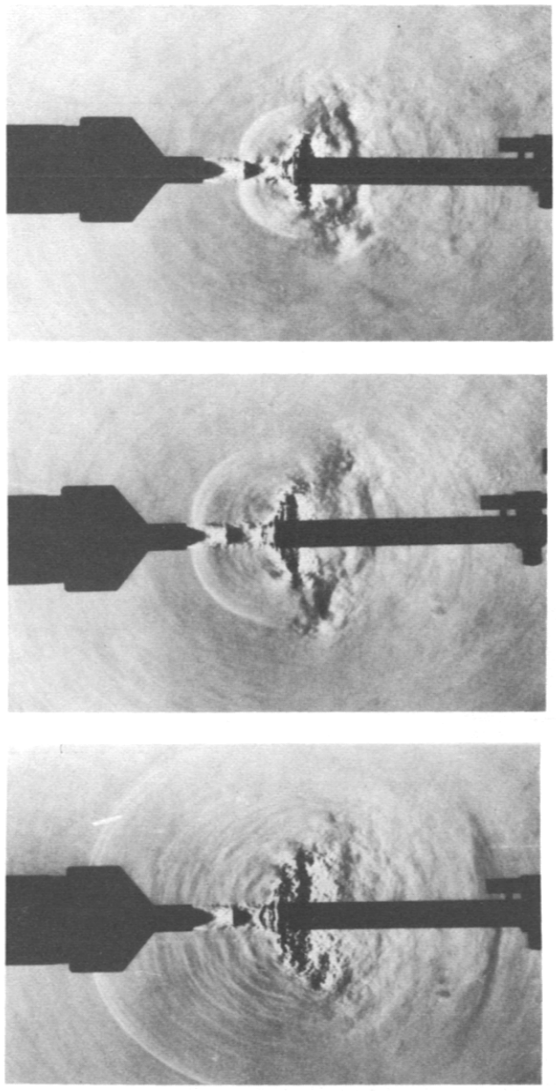


FIG. 62. External flow photographs for regurgitant oscillation mode showing the strong acoustical disturbance; flow direction is from left to right (from Ref. 66).

(Thompson[71]). However, some additional assumptions must be made to formulate the boundary conditions at the open end of the tube, during the outflow phase and transition (from outflow to inflow) phase. Thompson assumed matching of the jet impulse from the nozzle (J_n) and from the tube (J_t) for the outflow phase, and J_t equal to 80% of the nozzle jet stagnation pressure for the transition phase. Moreover, he assumed a constant entropy increase coefficient (α) for the internal flow accounting for the viscosity and heat exchange effects as well as shock wave presence. For some value of α, chosen by the numerical experiments, Thompson obtained good agreement of the calculated and measured pressure traces.

The boundary conditions used by Thompson[70] correspond only to one particular case of the external flow pattern. However, the external flow can have various structures which depend on the supply to the ambient pressure ratio, the nozzle-tube spacing and the tube inlet geometry. As an example, Figs 63 and 64 show sketches of the external flow patterns by

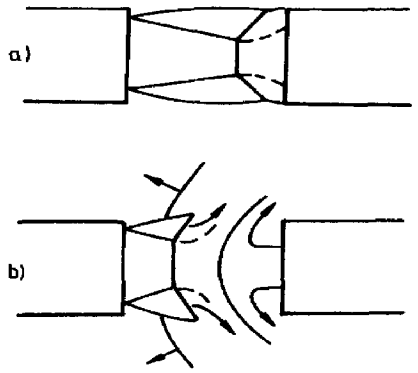


FIG. 63. Schemes of external flow patterns corresponding to inflow (a) and outflow (b) phases for regurgitant oscillation mode. Tube inlet placed in first cell of free jet.

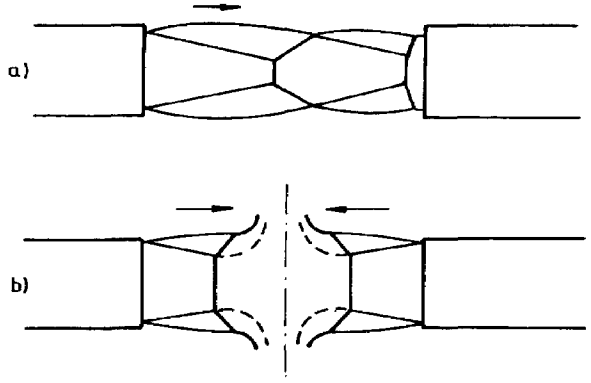


FIG. 64. Schemes of external flow patterns corresponding to inflow (a) and outflow (b) phases for regurgitant oscillation mode. Tube inlet placed in second cell of free jet.

Sobieraj and Szumowski[66] on the basis of flow photographs of the regurgitant oscillation mode for two tube spacings. In the first case the open end of the tube having thin walls was placed in the first cell of the free underexpanded jet from the nozzle. In this case the contact surface of both jets from the nozzle (J_n) and from the tube (J_t) during the outflow phase, is bowed downstream with $J_n > J_t$. In the other case (Fig. 64) which corresponds to a greater nozzle-tube spacing (the tube inlet placed in the second cell) the contact surface is planar. It confirms the condition $J_n = J_t$ assumed by Thompson[70] for this case.

The resonance tube flow corresponding roughly to the regurgitant oscillation mode was also analysed by Brocher *et al.*[9] They assumed a uniform jet from either a convergent or Laval nozzle under design conditions, impinging upon a resonator. The wave diagram of the tube flow (Fig. 65) used by Brocher *et al.*[9] displayed lines of compression and expansion waves considered as single disturbances of finite amplitudes. The authors assumed the Riemann variables to be maintained across the waves. In addition to this crude approximation of real flow Brocher *et al.*[9] assumed the following boundary conditions at the open tube end: the energy equation of steady adiabatic flow of the nozzle jet for the inflow phase and ambient pressure for the outflow phase. Under these assumptions Brocher *et al.* obtained a limit cycle shown in Fig. 66. In experiments made by Brocher *et al.* with uniform jets a cylindrical rod was inserted along the nozzle to diminish the total pressure in the central line of the nozzle jet. Consequently the contact surface of the nozzle and tube jets

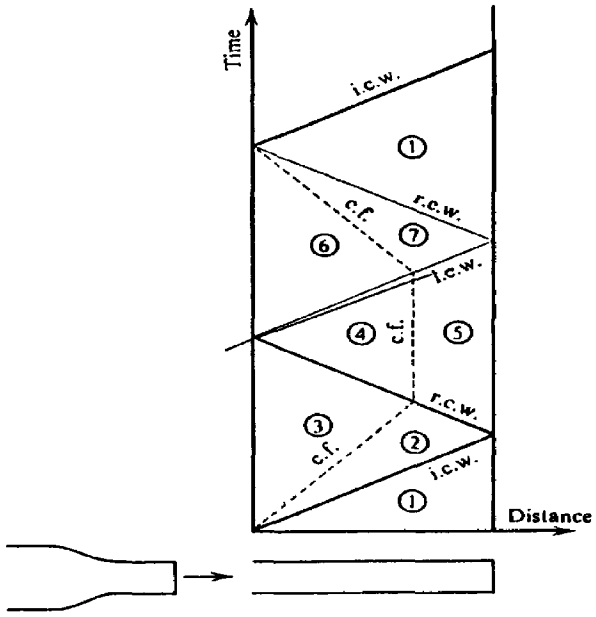


FIG. 65. Simplified wave diagram of the resonance tube (regurgitant mode); i.c.w.—incident compression wave, r.c.w.—reflected compression wave, i.e.w.—incident expansion wave, c.f.—contact surface front (from Ref. 9).

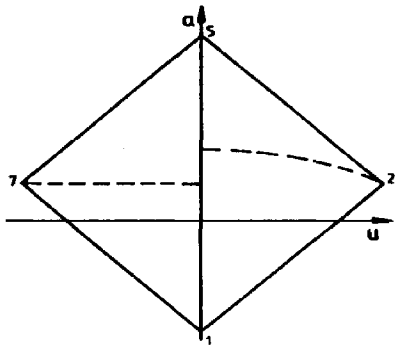


FIG. 66. Phase diagram for regurgitant oscillation mode; dashed lines show boundary conditions at the resonance tube inlet (from Ref. 9).

could be pushed further from the tube. In this way the tube emptied to a lower pressure and the flow oscillations were maintained.

5.2. The Screech Mode

In this oscillation mode an almost normal shock in front of the inlet of the resonance tube is present (Fig. 67). This shock oscillates at a high frequency. The presence of this oscillating shock wave, with a frequency as high as 20 kHz in this study, was the reason to call this the jet screech mode of resonance tube operation. The strength and location of this shock as well as its oscillation frequency depend upon the nozzle-tube spacing s/D, tube length L, nozzle pressure ratio p_o/p_a, etc. In contrast to the regurgitant modes in which the tube (or its greater part) is periodically discharged and swallowed, now only a very small amount of the

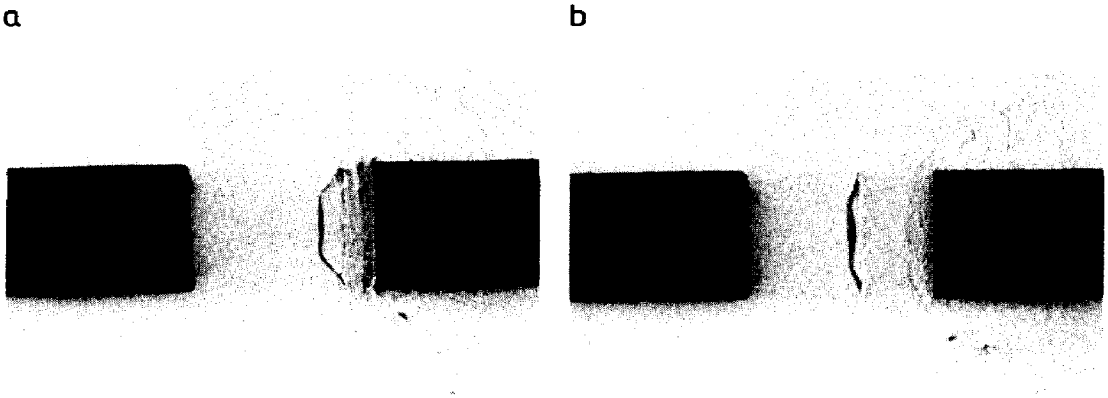


FIG. 67. External flow photographs for screech oscillation mode showing the extremal positions of shock wave; flow direction is from left to right. $d = 12$ mm, $L = 50$ mm, $p_0/p_a = 5.0$, fundamental frequency $f = 11.25$ kHz (from Ref. 66).

nozzle jet mass flow enters the tube. Hence, air remains in the tube during the screech mode oscillations. Thus, the normal shock wave in front of the tube inlet acts like a 'leaky piston' oscillating at high frequency. In consequence, the transition from regurgitant mode to screech mode is accompanied by a sudden increase of endwall gas temperature. Figure 68 shows the endwall temperature and the endwall pressure frequency variation, observed by Sahoria and Back,[61] when the supply to ambient pressure ratio was increased. It is evident that no significant endwall gas heating occurred in the jet regurgitant mode. In the latter mode, the frequency of the endwall pressure trace remained almost constant at the fundamental resonance frequency of the tube, which for the 76 cm tube was approximately 1000 Hz. As the resonance tube switched to the jet screech mode at a nozzle pressure ratio of 5.75, the endwall pressure signal indicated a jet screech frequency of about 3500 Hz. This was accompanied by a sudden jump in the endwall gas temperature to about 900 K, accompanied by charging of the resonance tube. As the nozzle pressure ratio was increased above 5.75, the screech frequency increased slowly until a nozzle pressure ratio of 7.85 was reached. The resonance tube flow then jumped to a higher stage of jet screech with an endwall pressure signal frequency of 18 kHz. This was accompanied by a sudden drop in the endwall gas temperature, which can be explained by a sudden decrease of the flow velocity amplitude. Measurements showed that large-amplitude oscillations of the shock wave in the jet flow were accompanied by the presence of locally heated node points indicating the presence of standing waves inside the tube.

The intense heating of the endwall gas in the first stage of the screech mode is explained by Sarohia and Back,[61] as follows:

(i) In a primary jet screech mode, the jet flow shock oscillation frequency is in tune with one of the resonance frequencies of the tube.

(ii) Large-amplitude oscillations of the normal shock between the nozzle exit and the tube inlet then occur with amplitudes as large as 15–20% of the nozzle exit diameter.

(iii) As compared to the inflow phase of the jet regurgitant mode, only a small fraction of the nozzle jet flow enters the tube. Consequently, little mass exchange results during each cycle of the jet screech mode.

(iv) Most of the dissipative heat energy generated by irreversible processes across the compression wavefronts is trapped between the node points of the tuned resonance tube system in this jet screech mode. Because of reduced mass exchange between the cold nozzle jet flow and the oscillating hot tube flow in each jet screech mode cycle, intense heating of the gas near the endwall results. This is due to cumulative heat addition to the trapped endwall gas by dissipative processes across the high-frequency compression waves. These originate from the screeching shock wave in the jet flow acting like a 'leaky piston'.

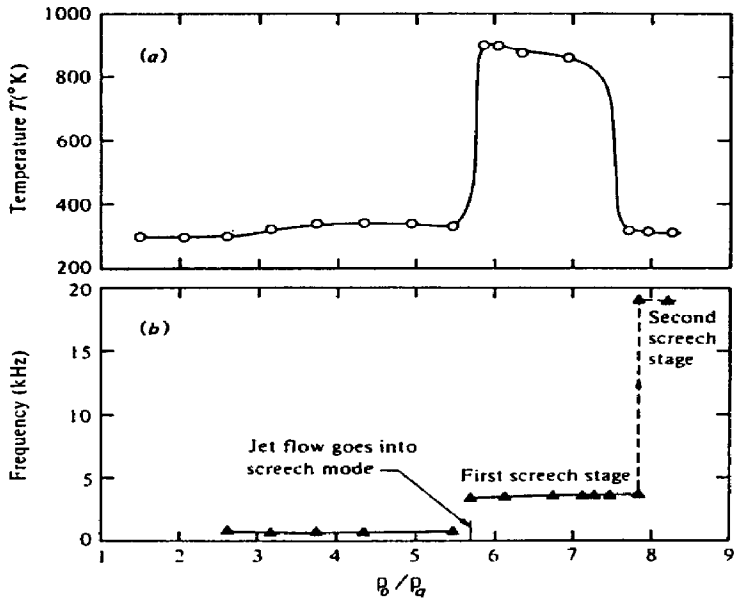


FIG. 68. Variations of endwall temperature and frequency during the screech mode oscillation with supply pressure (from Ref. 65).

(v) As the resonance tube flow goes out of resonance with the oscillating shock wave, e.g. with higher stages of the jet screech mode such as is shown in Fig. 68, no standing waves appear inside the tube and insignificant heating of the endwall gas occurs.

A few attempts were made to develop a theory describing the oscillation mechanism of screech mode, but none was satisfactory. Mørch,[46] assuming one-dimensional flow along the jet axis between a normal shock and a blunt body as a resonator, derived a linear wave equation and boundary conditions at the shock and at the resonator. As a solution he obtained several resonant frequencies corresponding to various amplification factors. The theory applies, however, to shock oscillation with a small amplitude only. Therefore, it may be adequate for the screech mode or possibly also for the regurgitant mode with a very short cavity. Kawahashi and Suzuki[30] considered the flow between the nozzle and the resonator, as well as an internal flow in the resonator. These flow fields were separated by a hypothetical, massless membrane. The authors applied one-dimensional linear theory and introduced some experimentally found jet properties. The results showed that the oscillations are self-excited owing to the negative acoustic resistance in certain ranges along the jet. In those unstable ranges, shock vibrations in the axial direction arise, with small amplitude and high frequency.

Some elucidation of the screech mode appears from investigations of the jet impinging upon a face of a rod. In this case oscillations of frequencies comparable to the screech mode frequencies occur. The flow photographs during the jet–rod interaction (Fig. 69) exhibit close similarity to those for the screech mode oscillation (Fig. 67). Powell,[48] noting this similarity (also observed in his experiments) postulates an oscillation mechanism for both the jet–rod interaction and the screech mode in the Hartmann generator, assuming the one-dimensional model of the flow in the space between the shock wave and the rod/resonance tube inlet (also Mørch[46]). Powell[48] suggested that the unstable shock wave which responds to pressure waves reflected back to it from the downstream face transforms the steady flow energy of the jet into the oscillation energy. In this respect, this hypothesis differs from Mørch's which involved an overdamped resonance supplemented by a postulated separate driving mechanism.

According to Powell's assumed model, the transit time for a particular phase of a pressure wave to traverse the region from the shock wave to the plate back again corresponds to about a quarter of the period of oscillation, and so

$$\frac{1}{4f} \approx \int_{s-s_h}^{s} \frac{dx}{a+u} - \int_{s}^{s-s_h} \frac{dx}{a-u} \tag{5.1}$$

where s_h is the shock spacing. Powell[48] made a rough estimate of these times by ignoring the change in position of the shock wave, taking s_h constant at its mean value. He also assumed that the axial velocity in the stagnation flow fell linearly to zero at the plate, ignoring any change in sound speed from the ambient value a_0 as well as ignoring the presence of a possible stagnation bubble. The average axial flow velocity will be less than this, so the transit time, and, therefore λ/D, will be overestimated on this account. Then, it followed that, if M_2 is the Mach number immediately downstream of the shock wave,

$$\frac{1}{4f} \approx \frac{s_h}{M_2 a_0} \ln \frac{1+M_2}{1-M_2}. \tag{5.2}$$

Powell[48] gave an example taking the mean value $s_h = 0.6\,D$, and the Mach number M_1 ahead of the shock wave as 1.99 (from the centreline Mach number distribution of his free jet). Then $M_2 = 0.58$, and he found $\lambda/D = 5.48$. The corresponding experimental value was 4.2. Hence, both the calculated and experimental values were of the same order in spite of the gross simplification of what is a very non-linear situation.

Further, the flow under consideration is not only non-linear but also far from one-dimensional. The flow photographs (Figs 67 and 69) show the shock wave to be spaced from the resonance tube inlet (Fig. 67) or from the rod (Fig. 69) a distance less than the jet diameter. Moreover, in the oscillation phase corresponding to the photographs on the left of both the figures the shock wave is strongly turned downstream at the jet rim. Thus, in this phase the jet downstream of the shock wave has a subsonic low velocity core surrounded by a high velocity (probably supersonic) layer. This pattern also occurred in the other extremal phase (the photographs on the right of Figs 67 and 69) but in this phase the supersonic layer, if it occurs, is narrow.

In view of this, the following mechanism of the screech oscillation mode can be suggested (Sobieraj and Szumowski[66]). The screech mode appears when the rod face (the tube inlet) is placed in the compression zone (let us consider the first cell) of the free jet. Then the cell is partly destroyed by the shock wave which occurs in the expansion zone of the cell. In one phase of the oscillation the shock wave is close to the rod (see photograph a in Fig. 69). In this phase the rod face pressure has a maximum value (Fig. 70). This pressure is associated

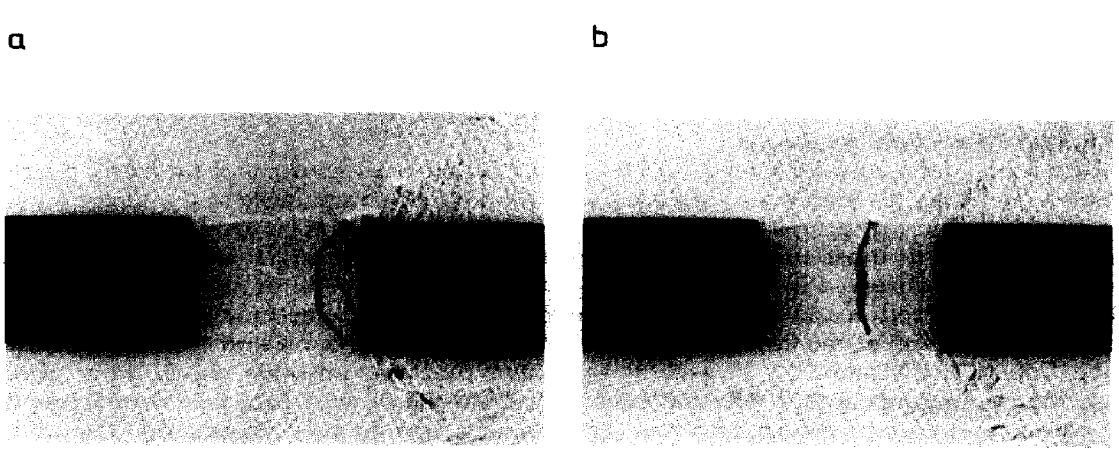


FIG. 69. Flow photographs for jet impinging upon a rod; flow direction is from left to right. $d = 12$ mm, $p_0/p_a = 5.0$, fundamental frequency $f = 13.2$ kHz (from Ref. 66).



FIG. 70. Pressure trace measured on the rod face (from Ref. 66).

with the strong centrifugal forces across the high velocity layer (surrounding the subsonic core) which curves substantially to flow around the rod. However, the rod face pressure is then too high to maintain the shock wave in the stable position because this position corresponds to the minimum stagnation pressure just behind the shock. (This situation is illustrated in Fig. 71, showing the centreline Pitot pressure distribution in the free jet and both sketches of the extremal flow patterns. Mach disc positions in these sketches indicate, at the Pitot pressure line, the stagnation pressures in the subsonic core.) Thus, the shock wave moves upstream; the Mach disc increases in diameter and the width of the high velocity layer decreases. Simultaneously, the centrifugal forces decrease due to the lower curvature of the high-velocity layer. Thus, the air in the stagnation region expands, causing the rod face pressure to decrease. This pressure has a minimum value (Fig. 70) in the phase corresponding to photograph b in Fig. 69. Now the rod face pressure is too low to sustain the shock wave far from the rod. Then the shock wave moves downstream and the cycle is repeated.

This mechanism explains also why the shock wave oscillations which are characteristic of the screech mode do not occur, when the shock appears in the compression zone of the cell. In this case a random increase of the rod face pressure causes the shock wave to shift to the new stable position corresponding to the lower Pitot pressure (the centreline Pitot pressure increases in the compression zone). When the resonance tube instead of the rod is used, the regurgitant oscillation occurs in this case. It can be concluded that the flow in the space between the shock wave and the rod head appears as the generator of the screech oscillation mode. When the rod is substituted for the resonance tube, the oscillations gain or lose their strength depending on the difference between the generator's natural frequency and the organ-pipe frequency of the resonance tube (Fig. 72).

5.3. Oscillation Ranges

Many experiments with the Hartmann–Sprenger generator enabled Smith and Powell[65] to find the ranges of both the regurgitant and screech modes of oscillation. These ranges are displayed in plots of the supply to ambient pressure ratio p_o/p_a versus the nozzle-tube spacing s/D representation (Fig. 73). With increasing s, for low pressure ratios, the screech mode appears as the first one when the tube inlet is present in the rear part of the first cell (compression zone; see Section 5.2). The screech mode transforms into the regurgitant mode when the tube inlet is approximately at the end of the first cell. Oscillations disappear for s corresponding to about half the length of the second cell. The screech mode is again evident if the tube inlet is shifted to the rear part of the second cell. The situation repeats with increasing s as long as the following cells are sufficiently distinct. For large pressure ratios the oscillations exist continuously for all nozzle-tube spacings. These observations were confirmed by the overall sound pressure level (OSPL) measurements (Sobieraj and Szumowski[66]) shown in Fig. 74. The OSPL traces have some depressions for low pressure ratios, corresponding to the oscillation free flow. For high pressure ratios the OSPL is nearly constant.

Sarohia and Back[61] focused their attention on the switch between the screech and the regurgitant mode. They conclude that the switch appears when the nozzle-tube spacing is approximately equal to the first free jet shock location. This agrees with Smith and Powell's[65] observations. The physical interpretation of the diagram in Fig. 73 is supported

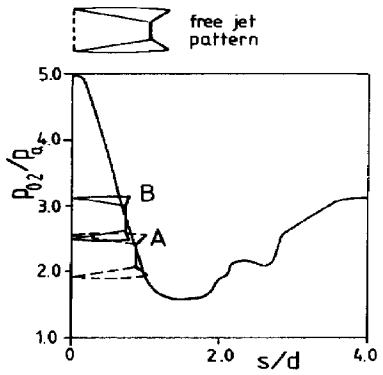


FIG. 71. Jet patterns ahead of the shock wave for the jet impinging upon a rod, synchronised with centreline Pitot pressure distribution along the free jet; pattern A—the shock nearest of the rod, pattern B—the shock farthest of the rod (from Ref. 66).

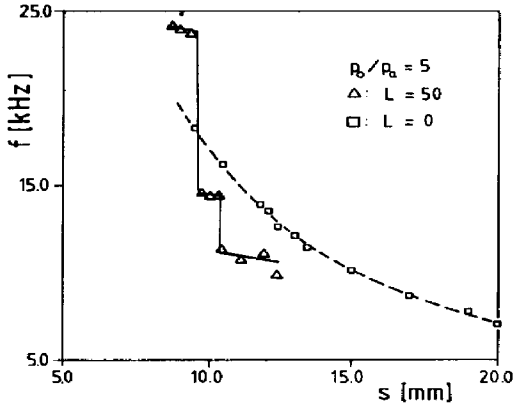


FIG. 72. Screech mode oscillation frequency for the jet impinging upon the tube (——) and the rod (– – –) (from Ref. 66).

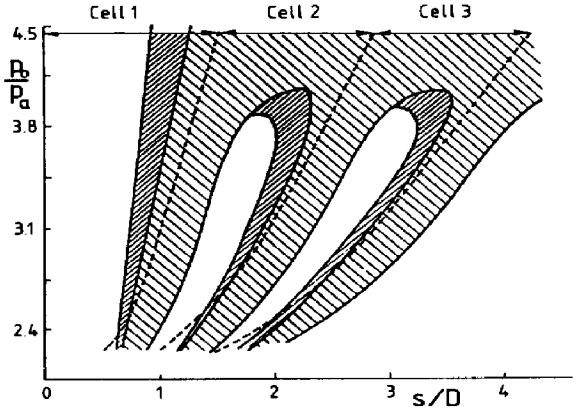


FIG. 73. Ranges of oscillation in Hartmann–Sprenger generator; thick and thin hatched areas correspond to screech mode and regurgitant mode, respectively (from Ref. 65).

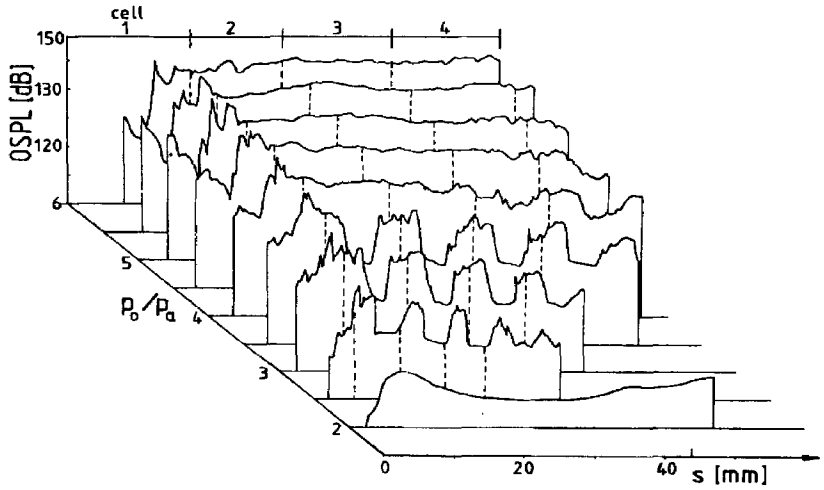


FIG. 74. Overall sound pressure level variations with increasing nozzle-tube spacing for some values of supply to ambient pressure ratio (from Ref. 66).

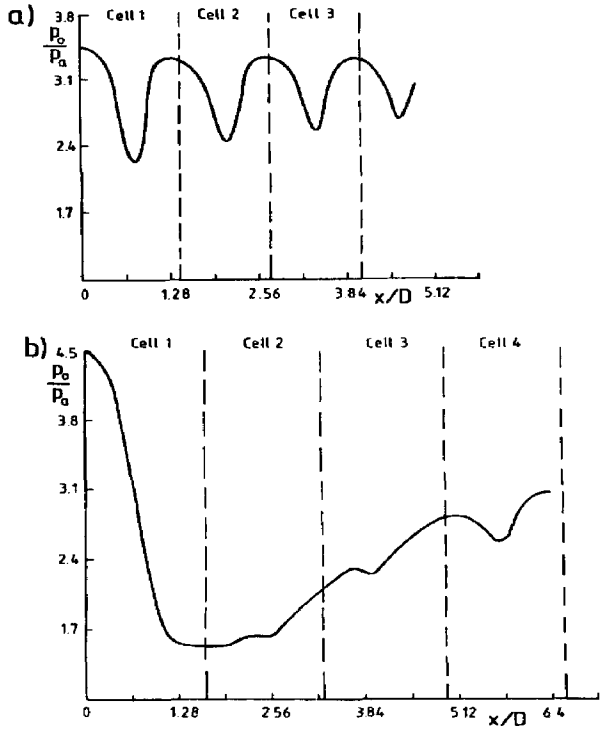


FIG. 75. Centreline Pitot pressure distributions along the free jet for two values of supply to ambient pressure ratio: (a) 3.45, (b) 4.5 (from Ref. 65).

by the centreline Pitot pressure distributions in a free jet shown in Fig. 75. For low pressure ratios (Fig. 75a) the Pitot pressure distribution has regular maxima and minima which correspond to the flow pattern displayed in Fig. 76a. In this case the screech mode appears when the shock wave upstream of the blunt body (rod or tube) in supersonic steady flow

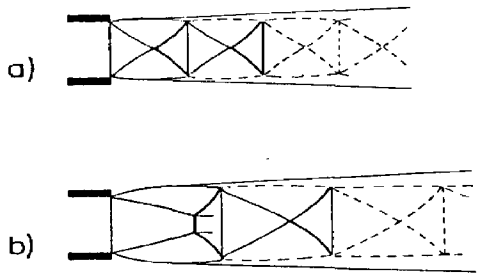


FIG. 76. Free jet structures; (a) supply to ambient pressure ratio $p_o/p_a < 4.0$; (b) $p_o/p_a > 4.0$.

would occur in the expansion zone of the cell, where the Pitot pressure decays. The regurgitant mode appears when the shock would occur in the region of increasing Pitot pressure. In those parts of the cells where the Pitot pressure is nearly constant no oscillations occur. For high pressure ratios (Figs 75b and 76b) the centreline Pitot pressure has a minimum at the end of the first cell. Thus, following the earlier considerations, one can conclude that the screech mode is possible in the first cell only. In the remaining cells the circumstances for the regurgitant mode occur.

6. CONCLUDING REMARKS

Though the oscillating transonic flows in the configurations considered (see Fig. 1) have been frequently studied in the literature, there are still many problems which remain unresolved, namely:

(i) Three-dimensional flow effects in curved ducts accompanied by separations, vortices and shock waves. (The investigations described in this review were concerned with two-dimensional effects only.)

(ii) Separation and reattachment behaviour in unsteady transonic flow. This is a key feature of oscillation mechanisms in transonic flow, especially in cases when the separation bubble extends into the receiver chamber. Many questions arise: How is the separation dependent on the shock wave Mach number? What is the cause of the splitting of the shock wave and the separation? How can we determine the separation velocity in a divergent channel?

(iii) Oscillating flow behaviour in a supersonic diffuser extended with a plenum chamber. The existing theoretical investigations concern the diffuser only; they assume an harmonic pressure excitation at the outlet cross-section.

(iv) Collar-nozzle flow oscillations in a slightly supersonic flow regime. The mechanisms of oscillation Modes I and II (see Figs 46, 53) were developed, in substance, for the subsonic flow range.

(v) The external boundary influence on the jet–cavity interaction mechanism was investigated for a simple configuration (Fig. 58a). However, a number of experiments indicate that the geometry of the nozzle and cavity lip, a plate placed at the nozzle outlet or at the cavity inlet, the presence of an external resonator and the like, influence considerably both the acoustic and thermal effects of oscillating flow. The mechanism of this phenomenon is not sufficiently elucidated. In particular, the role of the flow-acoustic coupling in this process is not clear.

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